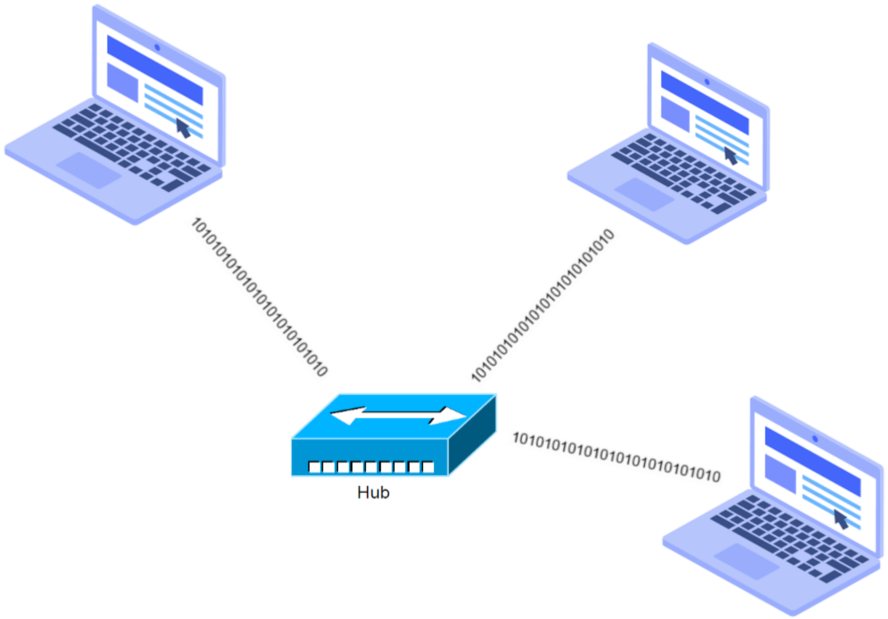
Network Fundamentals

Datalink layer and Ethernet

Physical layer

What the physical layer does

- Sending bits over a link



What the physical layer doesn't do

- Where does one frame end and the next one begin?
- How long is the frame
- Is the frame destined for me?
- Are there no bit errors in the frame?
- Deliver data to which upper layer?

Physical layer limitations

Layer 1 Limitations	Layer 2 Functions
Cannot communicate with upper layers	Connects to upper layers via Logical Link Control (LLC)
Cannot identify devices	Uses addressing schemes to identify devices
Only recognizes streams of bits	Uses frames to organize bits into groups
Cannot determine the source of a transmission when multiple devices are transmitting	Uses Media Access Control (MAC) to identify transmission sources

Ethernet – Layer 1 and Layer 2

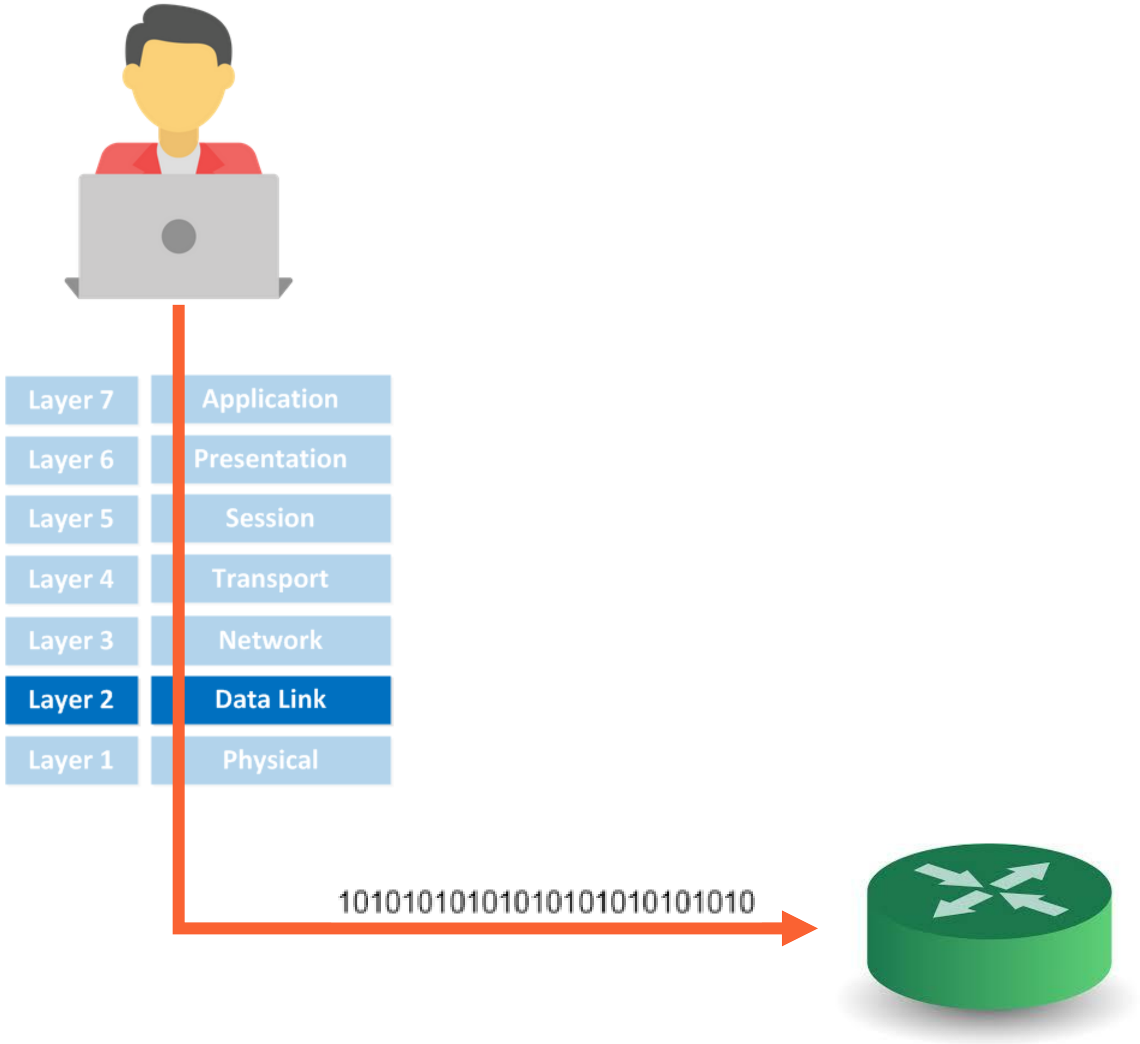
- **Ethernet** operates in the lower two layers of the OSI model: the Data Link layer and the Physical layer
  - Ethernet at **Layer 1** involves:
    - **Signals** = bit streams that travel on the media
    - **Physical components** that put signals on the media and various topologies

Ethernet Layer 1 performs a key role in the communication that takes place between devices, but each of its functions **has limitations**

The Data Link Layer when sending data

The data link layer performs these two basic services for sending data:

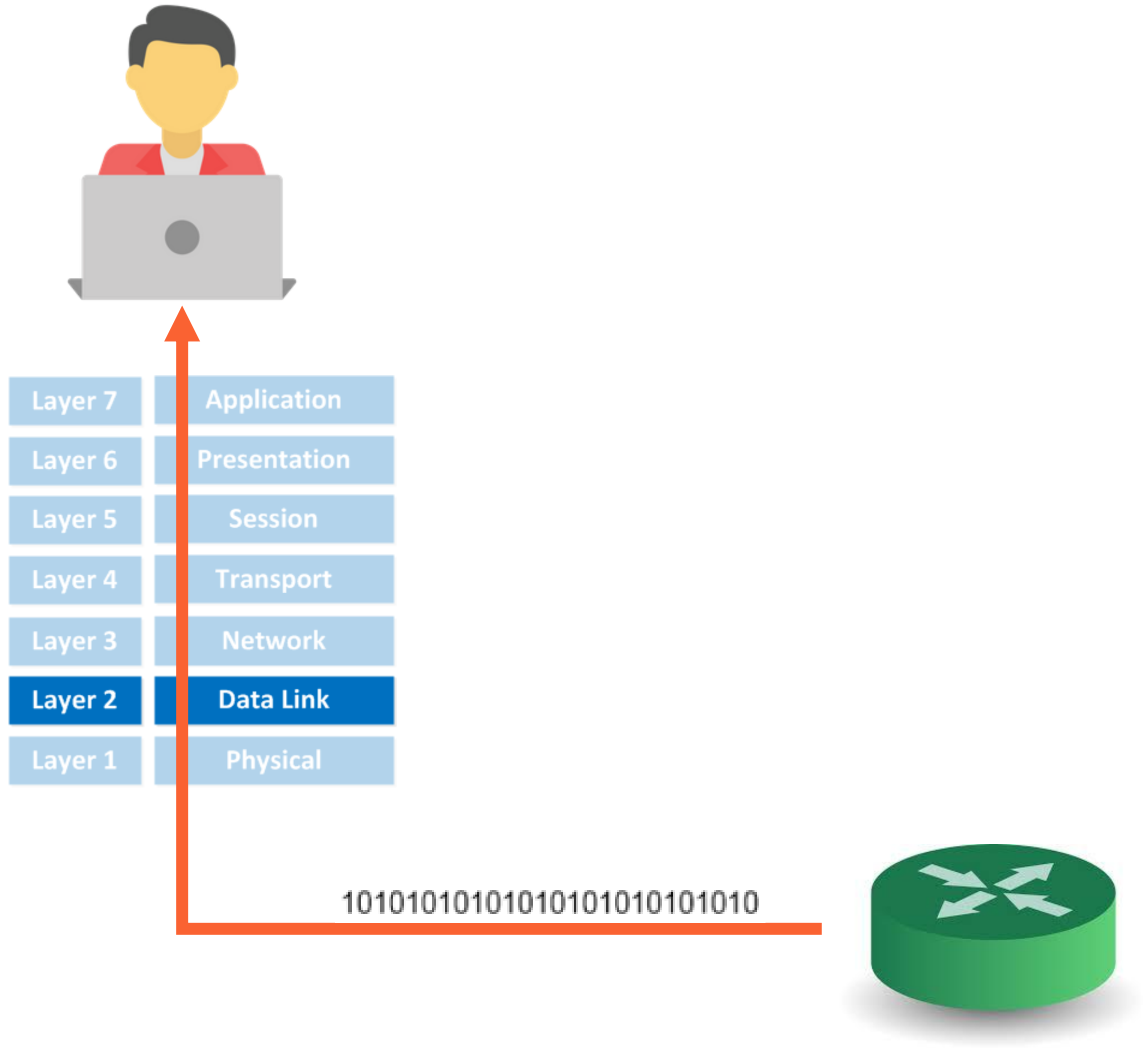
- It accepts Layer 3 packets and packages them into data units called **frames**.
- It controls media access control



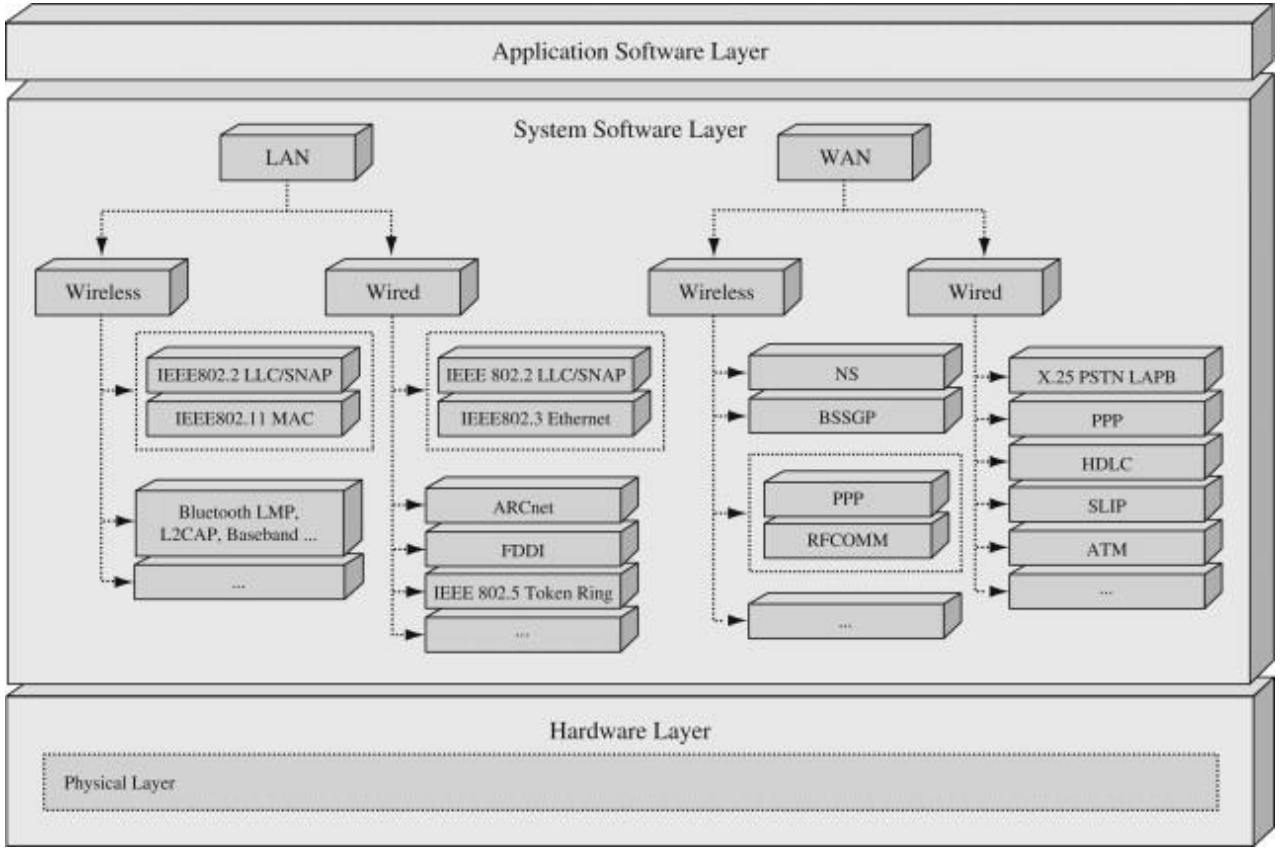
The Data Link Layer when receiving data

the data link layer performs these services for received data:

- It recognizes frames from a stream of bits
- It performs **error detection.**
- It determines the **destination** of the frame
- It delivers the packet to the correct Layer 3 protocol

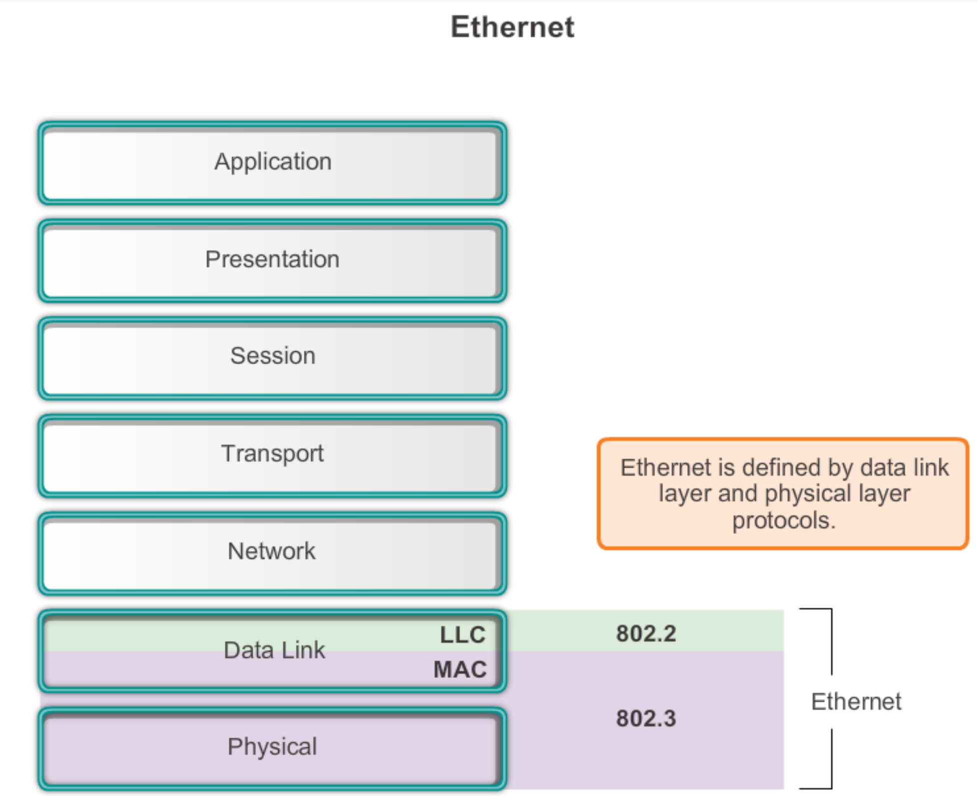


Datalink layer protocols



Ethernet – Standards and Implementation

- Ethernet operates in the lower two layers of the OSI model: the **Data Link** layer and the **Physical layer**.
- Most widely used LAN technology
- Family of networking technologies that are defined in the **IEEE 802.2 and 802.3** standards
- Supports data bandwidths of 10, 100, 1000, 10,000, 40,000, and 100,000 Mbps (100 Gbps)
- **an open standard**
- the IEEE 802.2 standard describes the LLC sublayer functions
- the IEEE 802.3 standard describes the MAC sublayer and the Physical layer

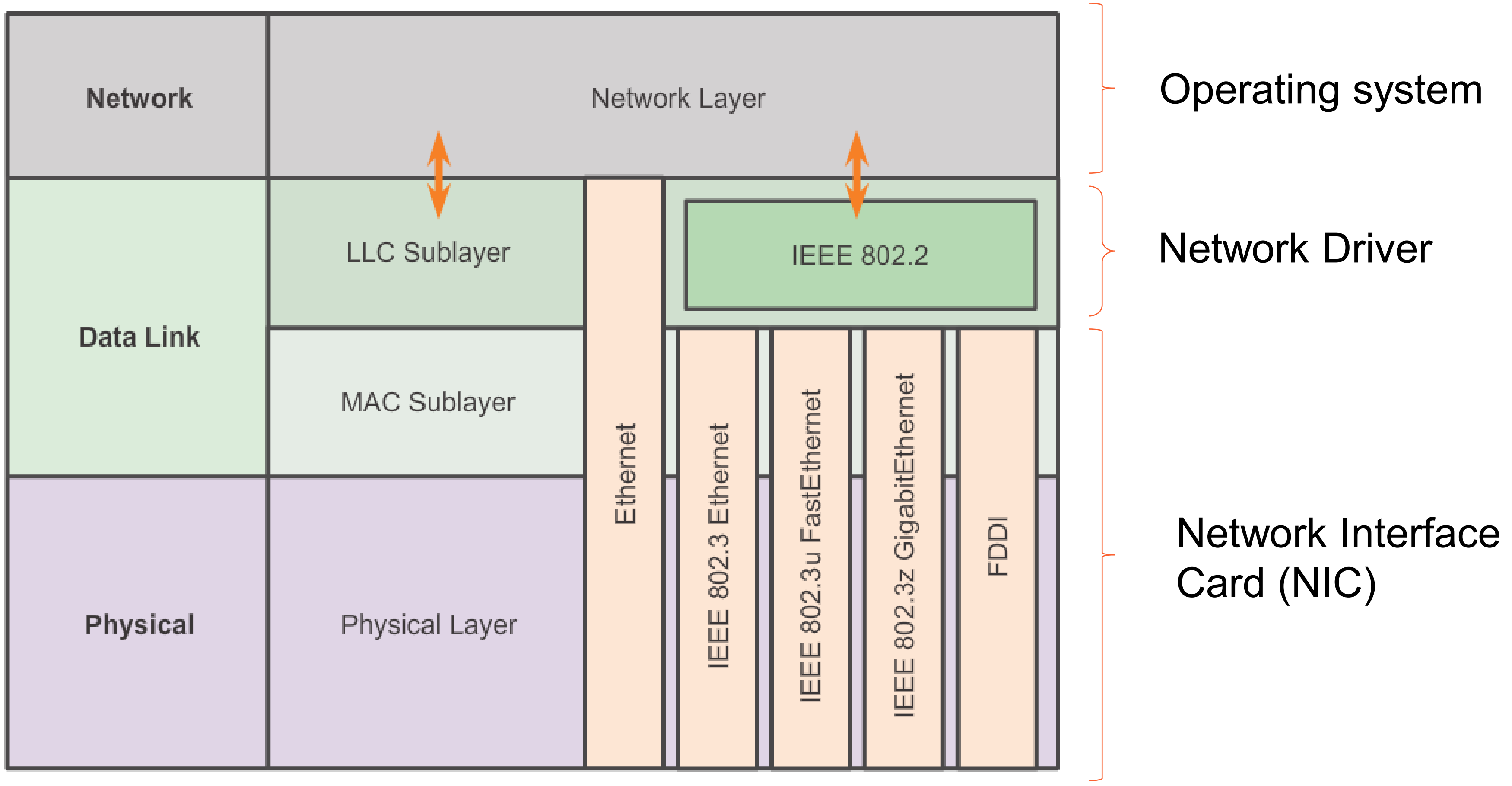


LLC and MAC Sublayers

Ethernet separates the functions of the Data Link layer into two distinct sublayers:

- the Logical Link Control (LLC) sublayer
- the Media Access Control (MAC) sublayer

LLC and MAC Sublayers



Logical Link Control Layer (LLC) sublayer

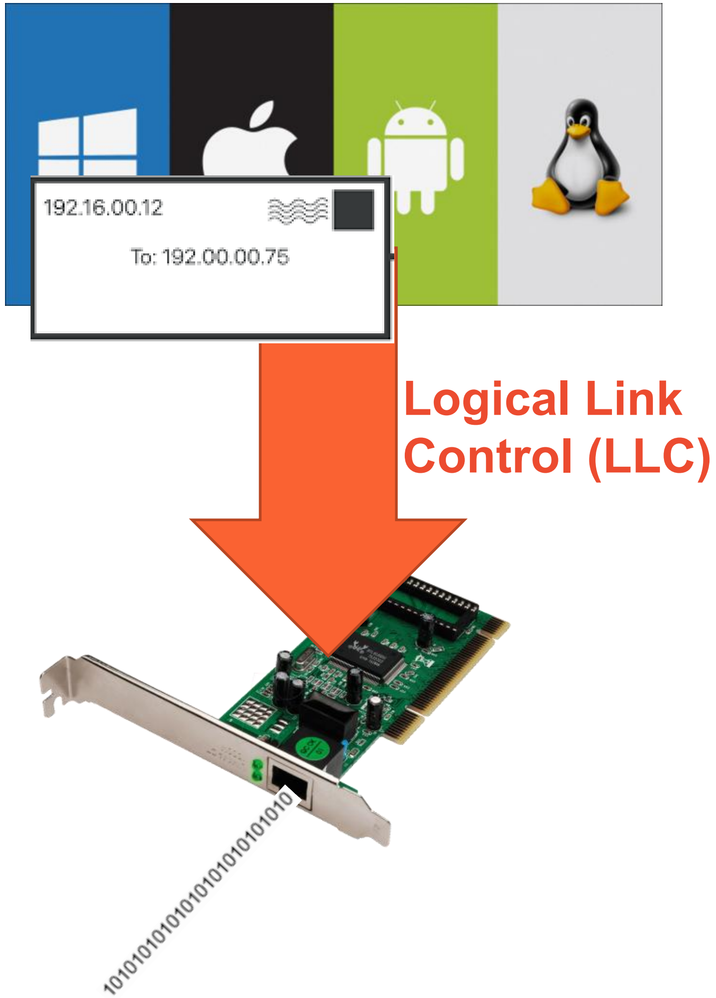
The upper Ethernet sublayer of the Data Link layer

Handles the communication between:

- the upper layers (**the networking software**)
- the lower layers (**typically the hardware**)

Takes the **network protocol** data, which is typically an **IP** packet, and adds control information to help deliver the packet to the destination node (frames the Network Layer packet)

- identifies the Network Layer protocol
- LLC communicates with the upper layers of the application and transitions the packet to the lower layers for delivery



Logical Link Control Layer (LLC) sublayer

The LLC is implemented in **software**

In a computer:
the LLC = **the driver software** for the Network Interface Card (NIC)

The NIC driver = a program that interacts directly with the hardware on the NIC to pass the data between the media and the Media Access Control sublayer

Logical Link Control (LLC)

- Makes the connection with the upper layers
- Frames the Network layer packet
- Identifies the Network layer protocol
- Remains relatively independent of the physical equipment

Media Access Control (MAC) sublayer

The lower Ethernet sublayer of the Data Link layer

- Media Access Control is **implemented by hardware**, typically in the computer **Network Interface Card** (NIC).
- The Ethernet MAC sublayer has **two primary responsibilities:**
  - **Data Encapsulation**
  - **Media Access Control**

Media Access Control (MAC) sublayer

Data encapsulation

- Frame assembly before transmission and frame disassembly upon reception of a frame
- MAC layer adds a header and trailer to the network layer PDU

Provides three primary functions:

- Frame delimiting (the 8-bit value marking the end of the preamble of an Ethernet frame)
- Addressing
- Error detection

Media Access Control (MAC) sublayer

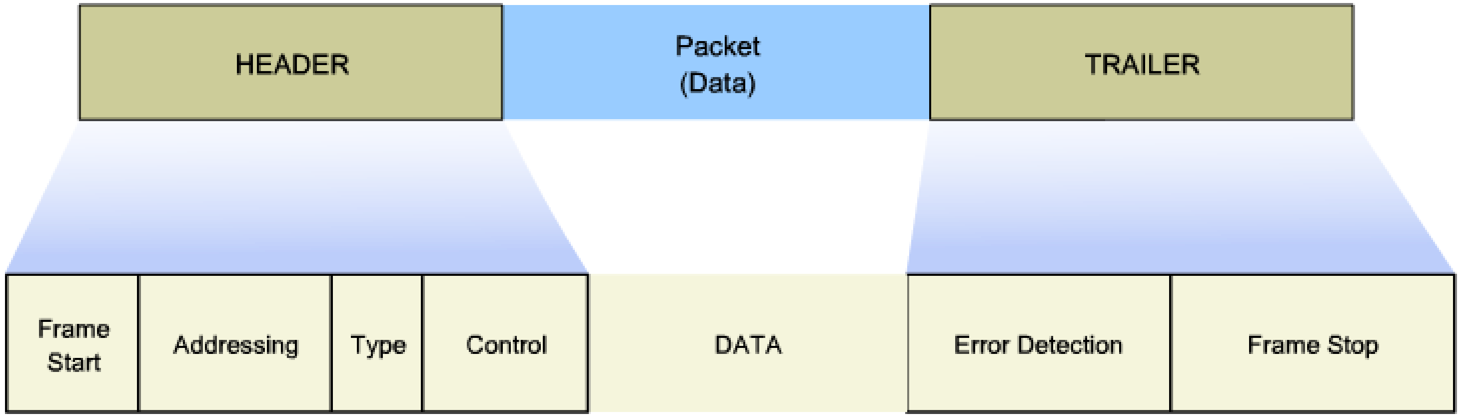
Media Access Control

- Responsible for the **placement of frames on the medium** and the removal of frames from the media
- Communicates directly with the physical layer
- If multiple devices on a single medium attempt to forward data simultaneously, the data will collide resulting in corrupted, unusable data
- **Ethernet** provides a method for controlling how the nodes share access through the use a Carrier Sense Multiple Access (CSMA) technology

Creating a Frame

generic frame field types include:

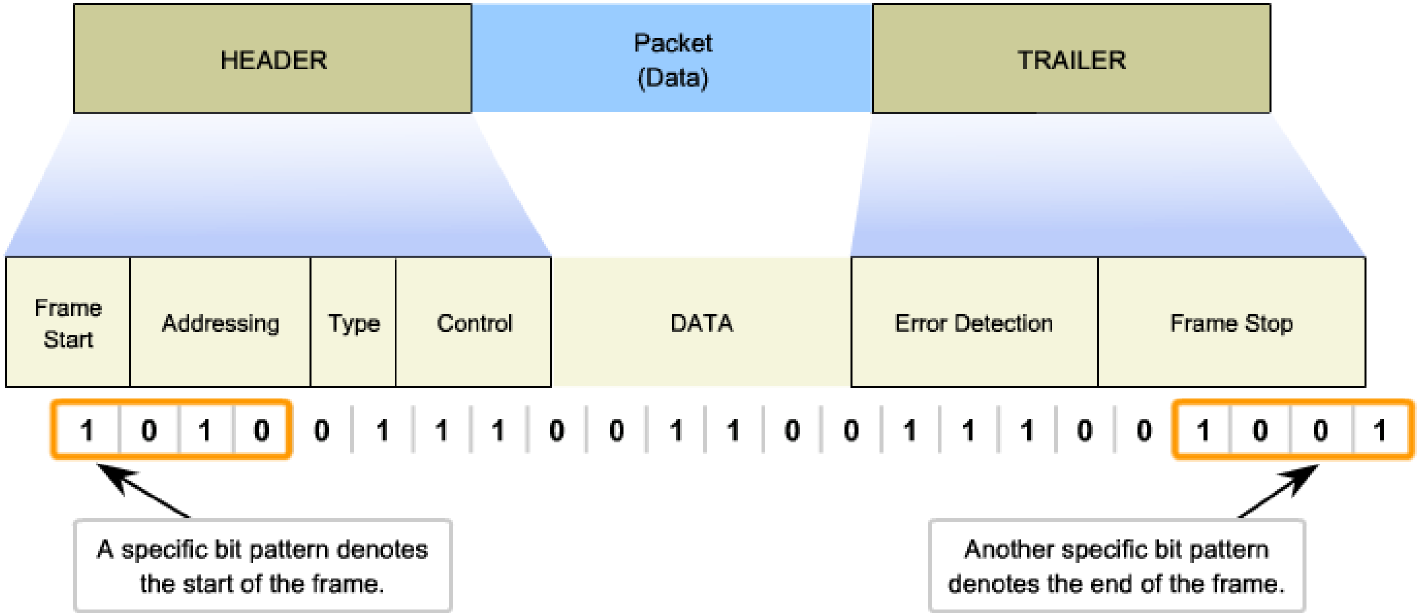
- **Frame start and stop indicator flags**: Used by the MAC sublayer to identify the beginning and end limits of the frame.
- **Addressing**: Used by the MAC sublayer to identify the source and destination nodes.
- **Type**: Used by the LLC to identify the Layer 3 protocol. (IPv4 or IPv6 in most cases)
- **Control**: Identifies special flow control services.
- **Data**: Contains the frame payload (i.e., packet header, segment header, and the data).
- **Error Detection**: Included after the data to form the trailer, these frame fields are used for error detection.



Layer 2 Frame Structure

the data link layer frame includes:

- **Header**: Contains control information, such as addressing, and is located at the beginning of the PDU.
- **Data**: Contains the IP header, transport layer header, and application data.
- **Trailer**: Contains control information for error detection added to the end of the PDU.



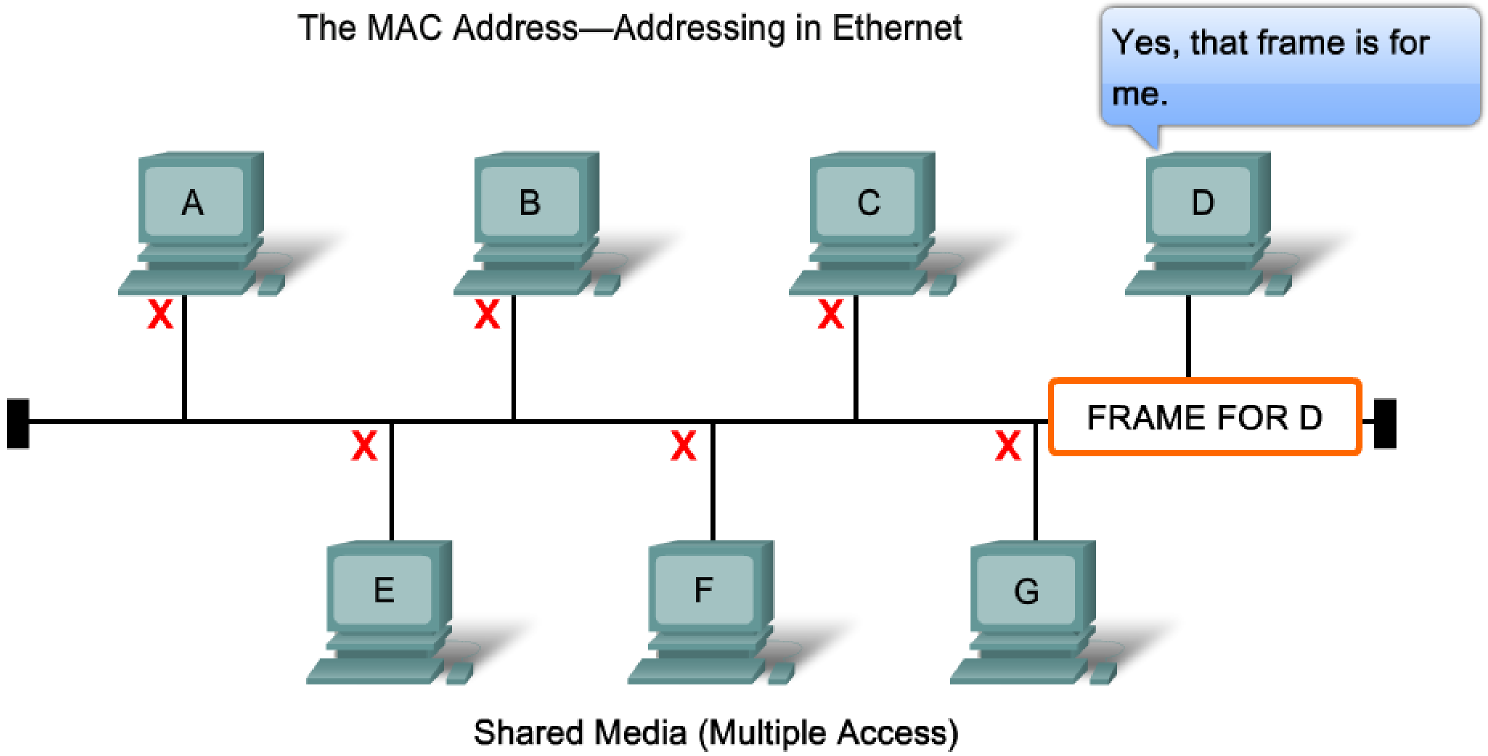
MAC Address: Ethernet Identity

- Initially : Ethernet = a bus topology
  - Every network device was connected to the same, shared media
  - In low traffic or small networks, this was an acceptable deployment
  - The main problem : **How to identify each device?**
    The signal could be sent to every device, but how would each device identify if it were the receiver of the message?

- Media Access Control (**MAC**) address was created to determine the source and destination address within an Ethernet network

- MAC addressing is added as part of a Layer 2 PDU
  An Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits

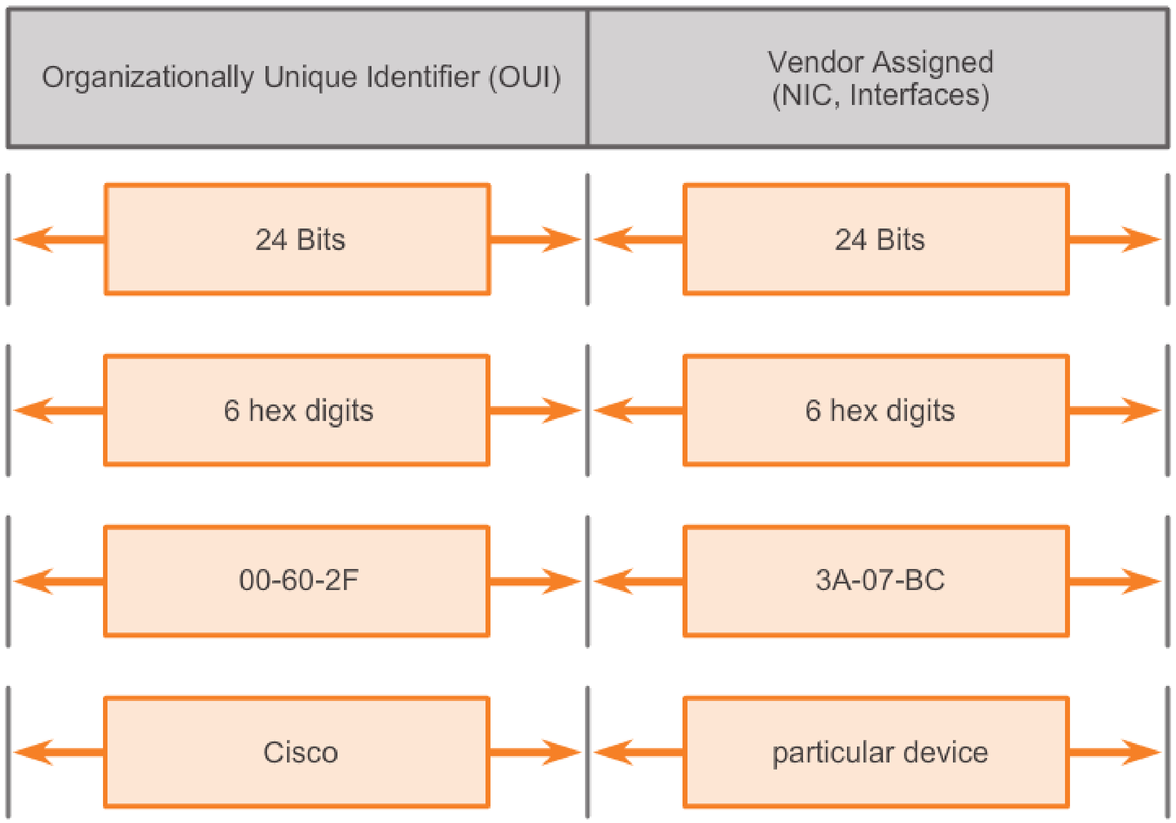
MAC Address: Ethernet Identity



All Ethernet nodes share the media.
To receive the data sent to it, each node needs a unique address.

MAC Address: Ethernet Identity

- MAC Address is 48 bits long
- 12 Hexadecimal numbers (0-16 = 0 – F)
- 24 first bits = OUI = unique provided to vendor
- 24 last bist = assigned per device by vendor



MAC Addresses and Hexadecimal

Decimal and Binary equivalents of 0 to F Hexadecimal

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Selected Decimal, Binary and Hexadecimal equivalents

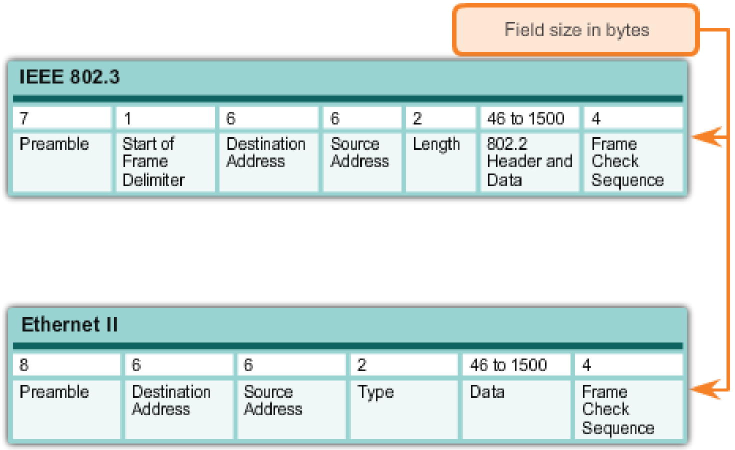
Decimal	Binary	Hexadecimal
0	0000 0000	00
1	0000 0001	01
2	0000 0010	02
3	0000 0011	03
4	0000 0100	04
5	0000 0101	05
6	0000 0110	06
7	0000 0111	07
8	0000 1000	08
10	0000 1010	0A
15	0000 1111	0F
16	0001 0000	10
32	0010 0000	20
64	0100 0000	40
128	1000 0000	80
192	1100 0000	C0
202	1100 1010	CA
240	1111 0000	F0
255	1111 1111	FF

Frame Processing

- The MAC address
  = a burned-in address **(BIA)** because it is burned into ROM (Read-Only Memory) on the NIC
  → meaning that the address is **encoded into the ROM chip permanently** - it cannot be changed by software
- When the computer starts up
  → the NIC copies the address into RAM
  → When examining frames, it is the address in the RAM that is used as the source address to compare with the destination address
- Each NIC views information to see if the destination MAC address in the frame matches the device's physical MAC address stored in RAM
  → **No match, the device discards the frame**
  → **Matches** the destination MAC of the frame, the **NIC passes the frame up the OSI layers**, where the de-encapsulation process takes place
- Example MACs: 00-05-9A-3C-78-00, 00:05:9A:3C:78:00, or 0005.9A3C.7800.
- Forwarded message to an Ethernet network, attaches header information to the packet, contains the source and destination MAC address

Ethernet Encapsulation

Ethernet frame structure adds **headers and trailers** around the Layer 3 PDU to encapsulate the message being sent
Can operate at 10 Gigabits per second and faster



Unicast MAC Address

Unicast MAC address

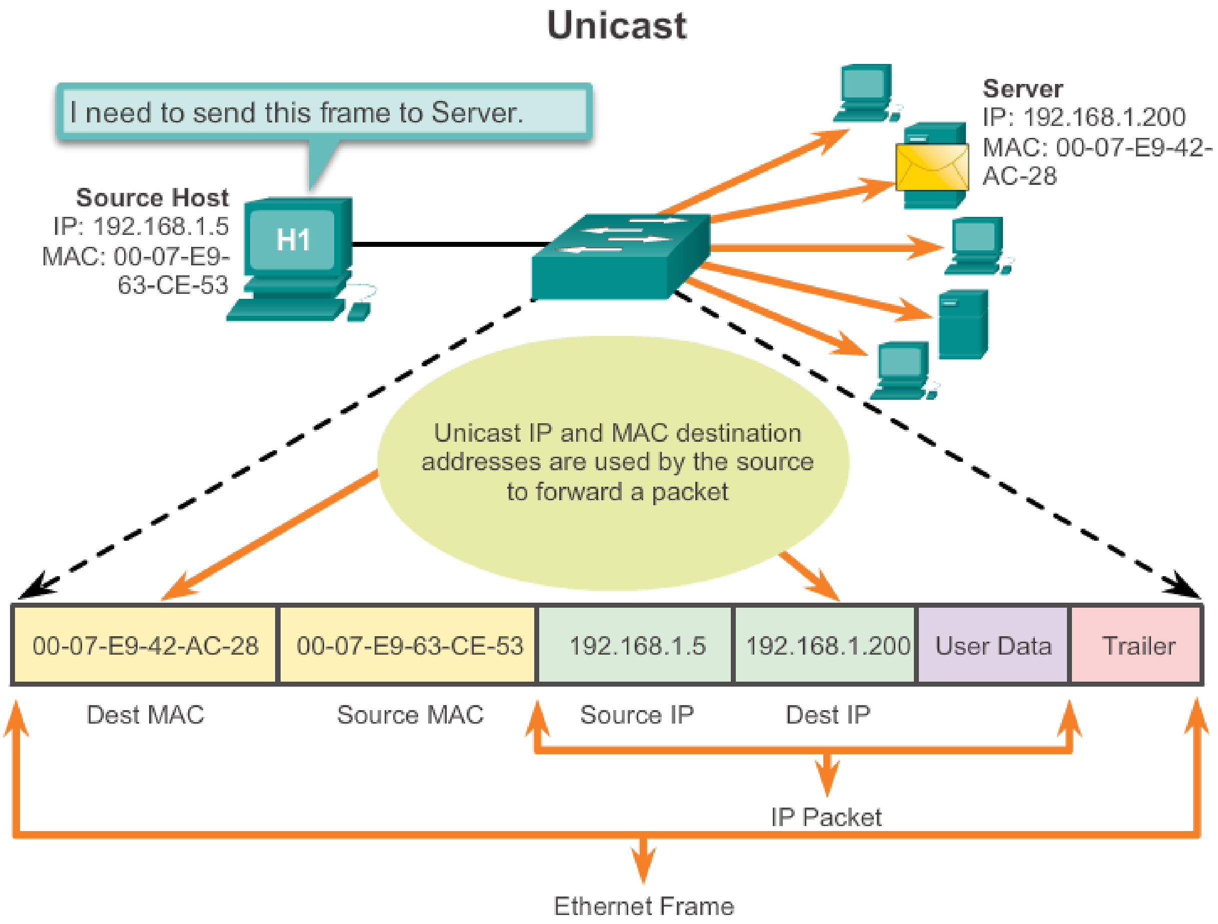
<u>Example:</u>

a host with IP address 192.168.1.5 (source) sends a frame to the server at IP addresses 192.168.1.200

therefore a destination IP address must be in the IP packet header

A corresponding destination MAC address must also be present in the ethernet frame header

the IP address and MAC address combine to deliver data to one specific destination host



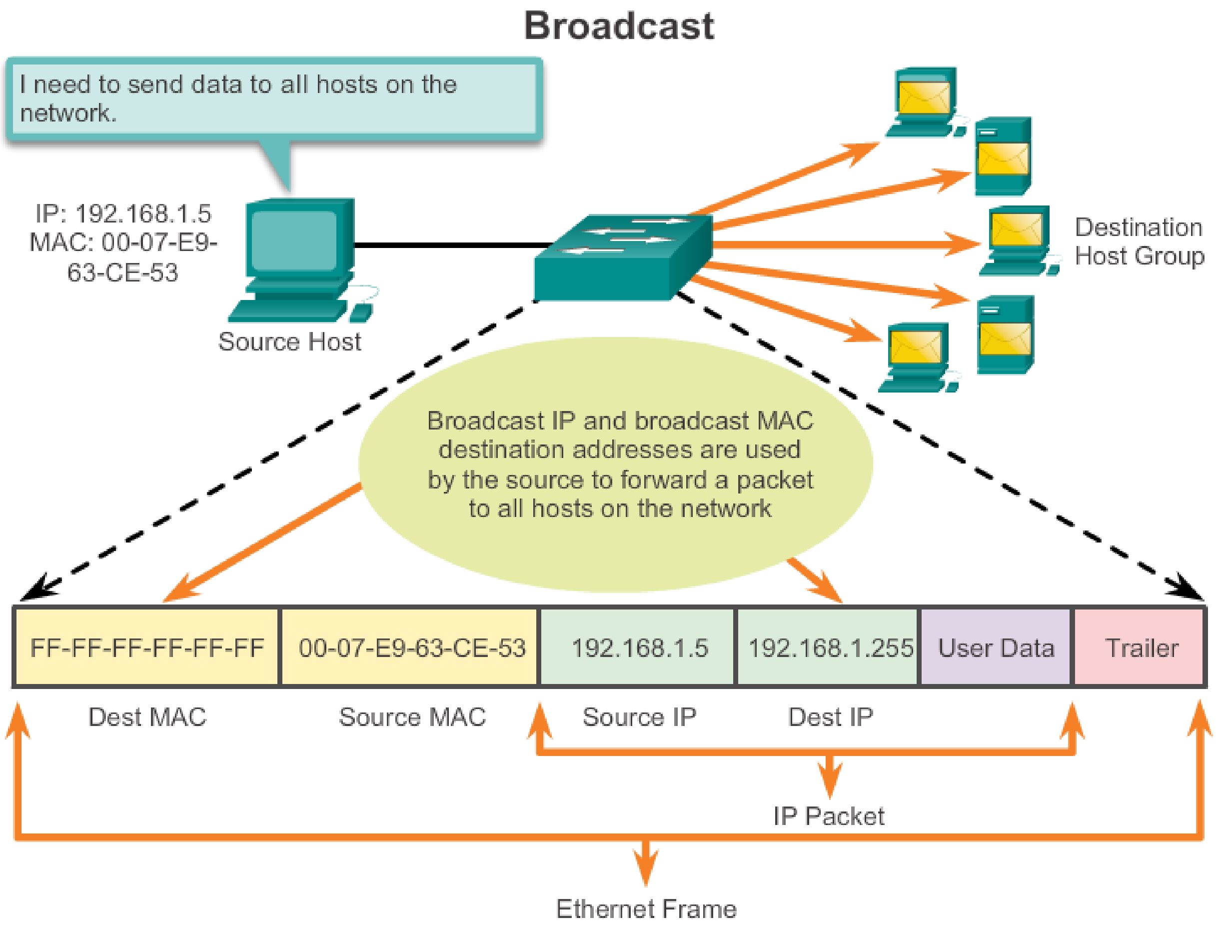
Broadcast MAC Address

Broadcast MAC address

<u>Example:</u>

a broadcast IP address for a network needs a corresponding broadcast MAC address in the Ethernet frame

On Ethernet networks, the broadcast MAC address is 48 ones displayed as Hexadecimal FF-FF-FF-FF-FF-FF



Multicast MAC Address

Multicast MAC address

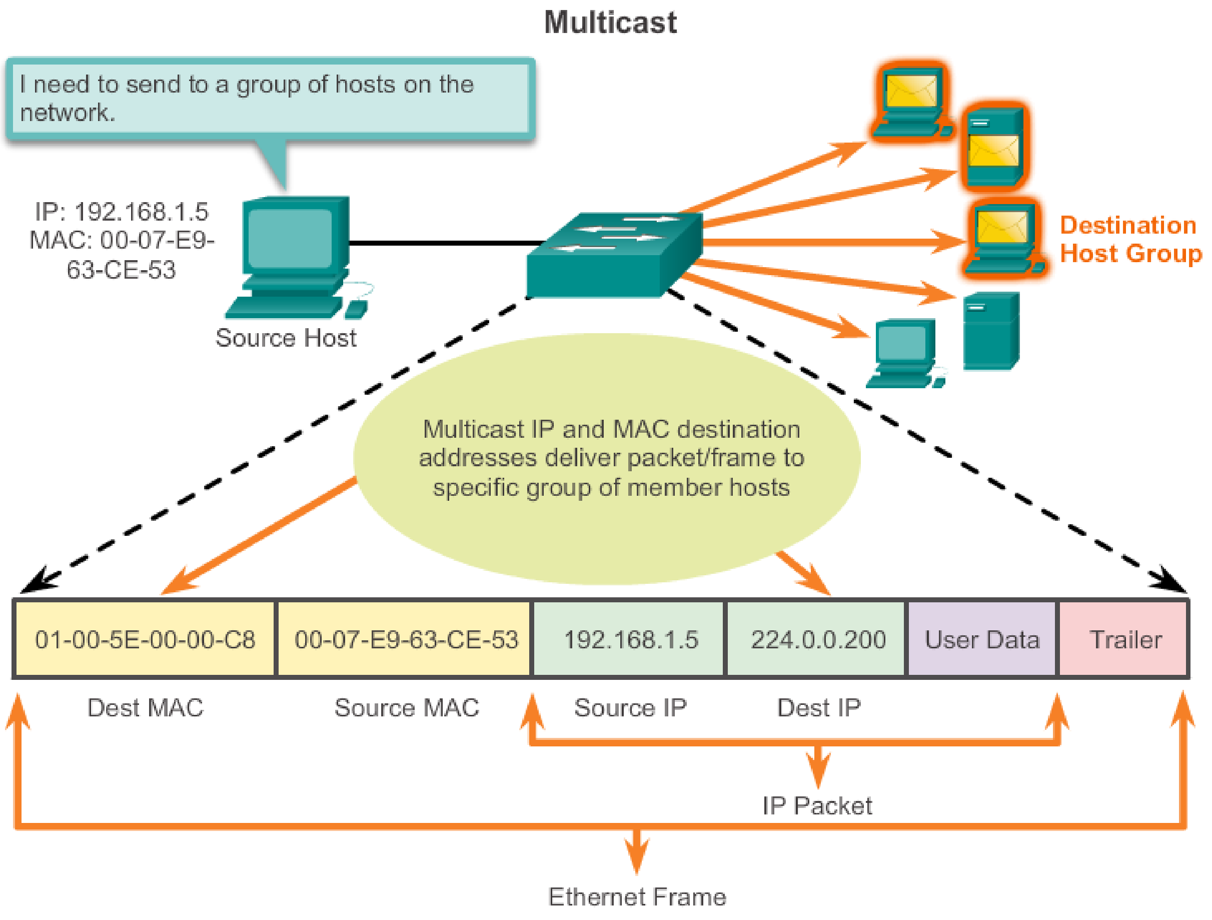
Multicast IP address : 224.0.0.0 tot en met 239.255.255.255

Example:

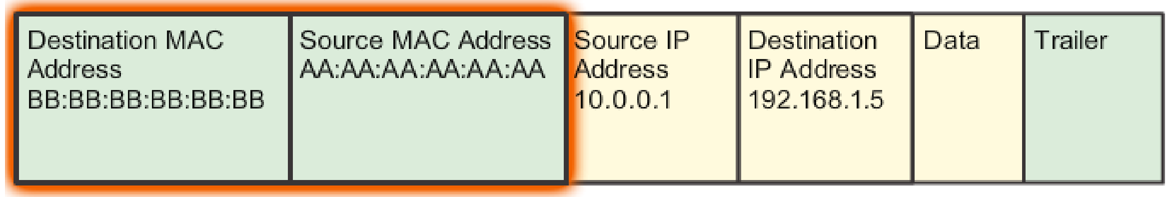
The multicast IP address requires also a corresponding multicast MAC address to deliver frameson a local network

The multicast MAC address is a special value that begins with 01-00-5E in hexadecimal

The value ends by converting the lower 23 bits of the IP multicast group address into the remaining 6 hexadecimal characters of the Ethernet address The remaining bit in the MAC address is always a "0"



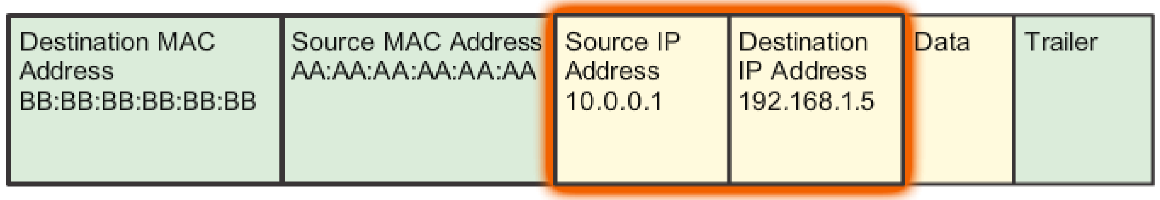
MAC VS IP



A switch examines MAC addresses.

MAC address

- This address **does not change** (BIA burn in address in ROM)
- Similar to the <u>name of a person</u>
- Known as **physical address** because physically assigned to the host NIC



A router examines IP addresses.

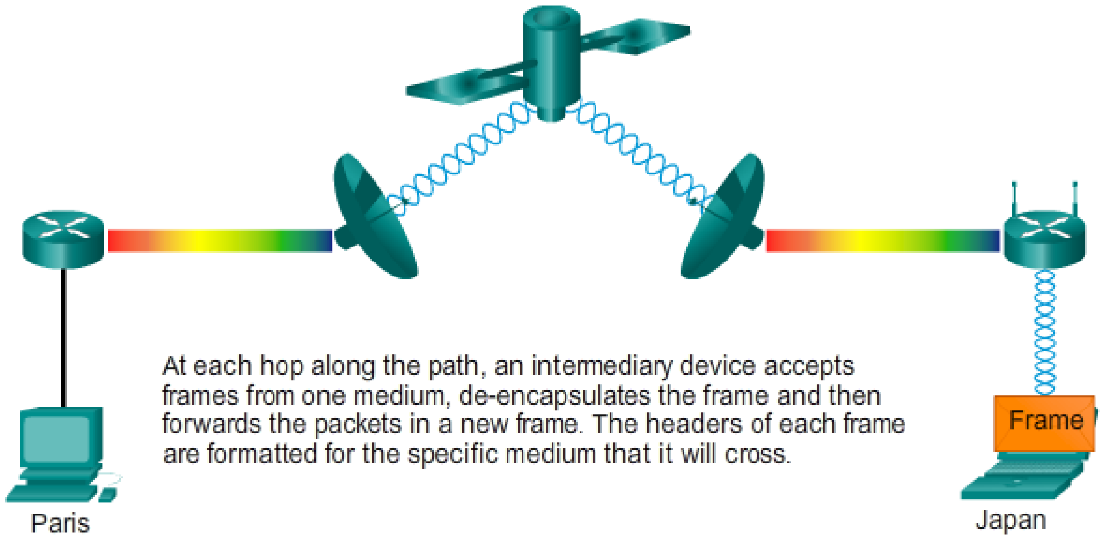
IP address

- Similar to the <u>address of a person</u>
- Based on where the host is actually located
- Known as a **logical address** because assigned logically
- **Assigned** to each host **by a network administrator**

Both the physical MAC and logical IP addresses are required for a computer to communicate just like both the name and address of a person are required to send a letter

Media Access Control

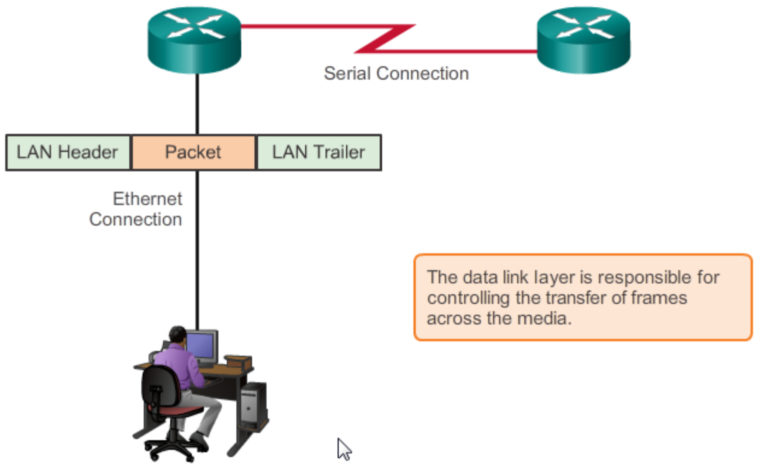
- As packets travel from source host Paris to destination host Japan, over different physical networks.
- These physical networks can consist of different types of physical media such as copper wires, optical fibers, wireless, and satellite links.
- The packets do not have a way to directly access these different media. It is the role of the OSI data link layer to prepare network layer packets for transmission and to control access to the physical media.

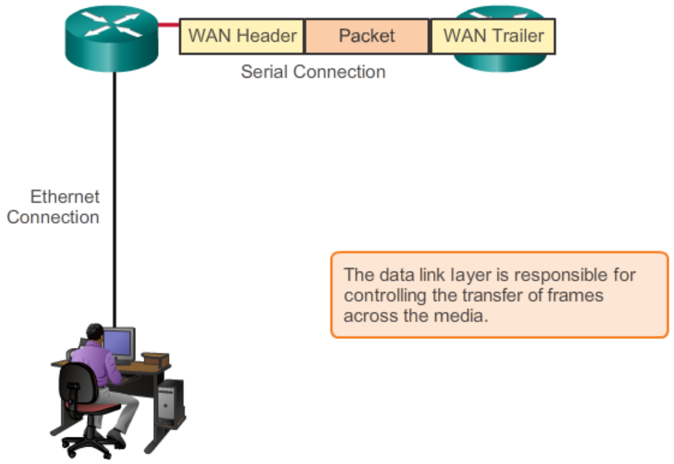


Providing Access to Media

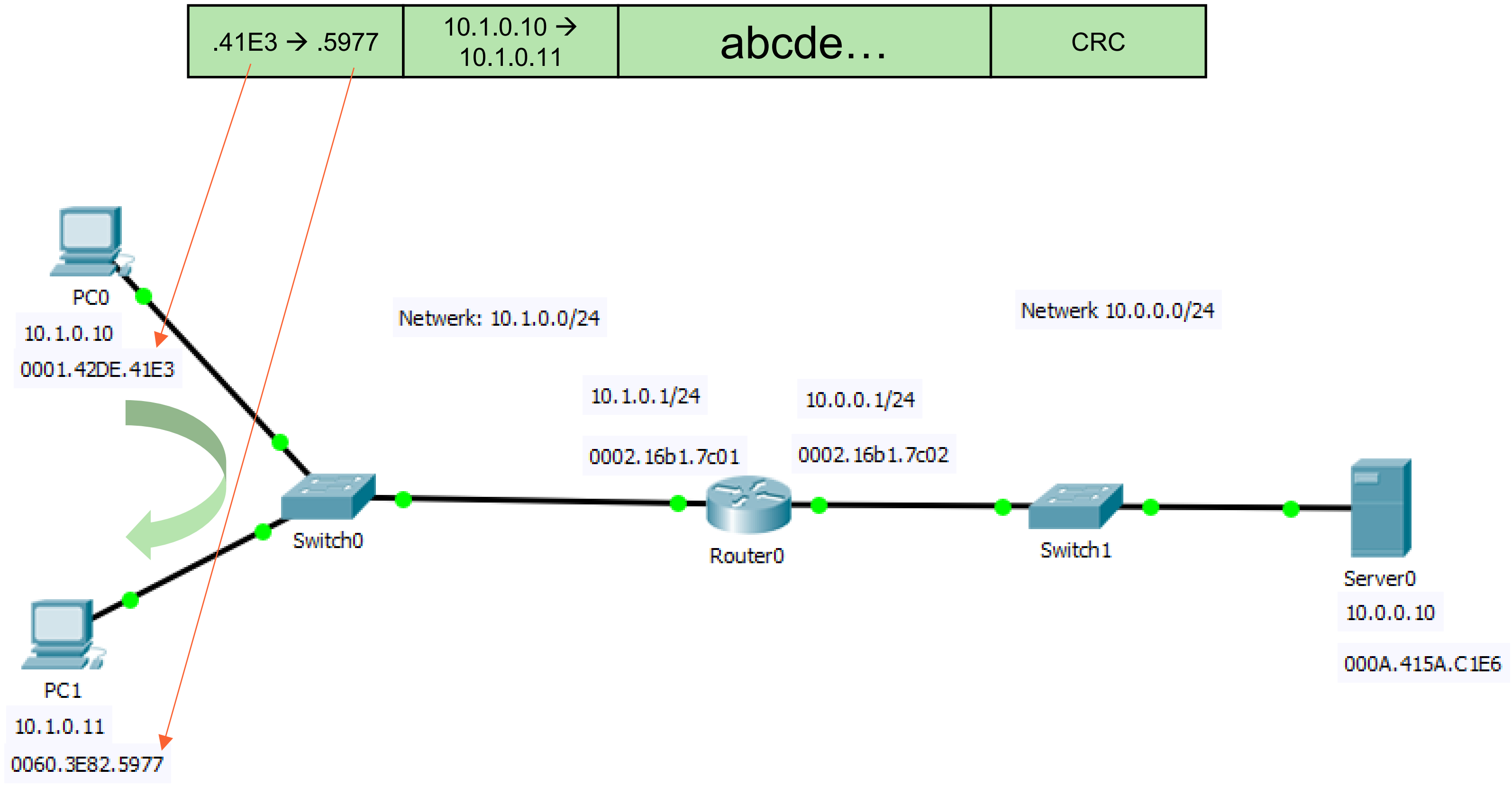
At each hop along the path, a router:

- Accepts a frame from a medium
- De-encapsulates the frame
- Re-encapsulates the packet into a new frame
- Forwards the new frame appropriate to the medium





Ping 10.1.0.10

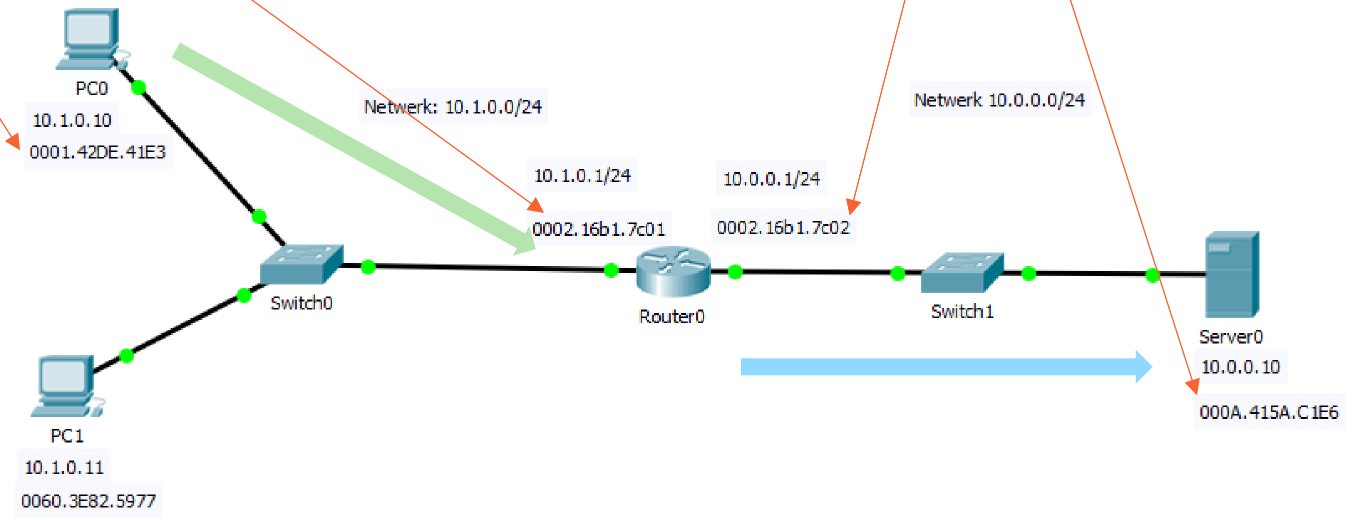


.41E3 → .5977
10.1.0.10 → 10.1.0.11
abcde…
CRC
PC0
10.1.0.10
0001.42DE.41E3
Netwerk: 10.1.0.0/24
10.1.0.1/24
0002.16b1.7c01
10.0.0.1/24
0002.16b1.7c02
Netwerk 10.0.0.0/24
Switch0
Router0
Switch1
PC1
10.1.0.11
0060.3E82.5977
Server0
10.0.0.10
000A.415A.C1E6

Ping 10.1.0.10

.41E3 → .7C01	10.1.0.10 → 10.0.0.10	abcde…	CRC

.7C02 → .C1E6	10.1.0.10 → 10.0.0.10	abcde…	CRC



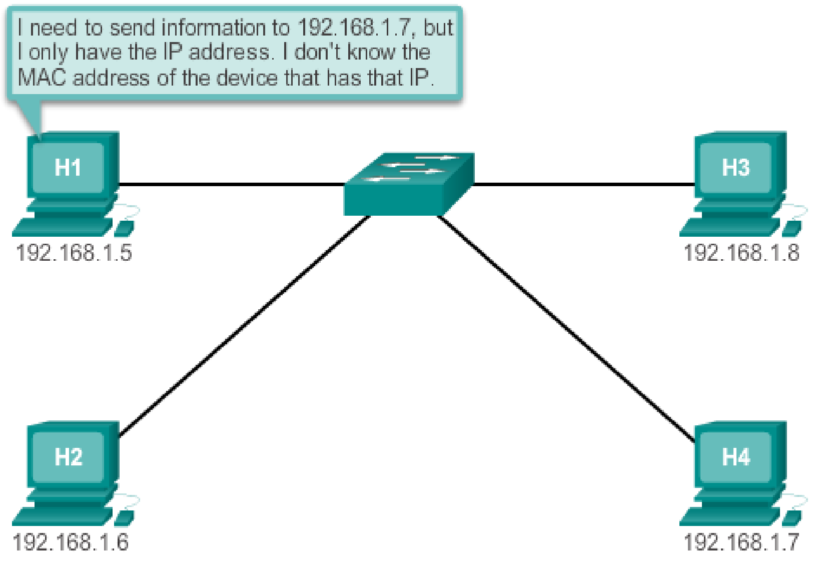
Introduction to ARP

ARP Purpose:

- To put a frame on the LAN network, a destination MAC address should be added during encapsulation
- What MAC address should be used for a node?

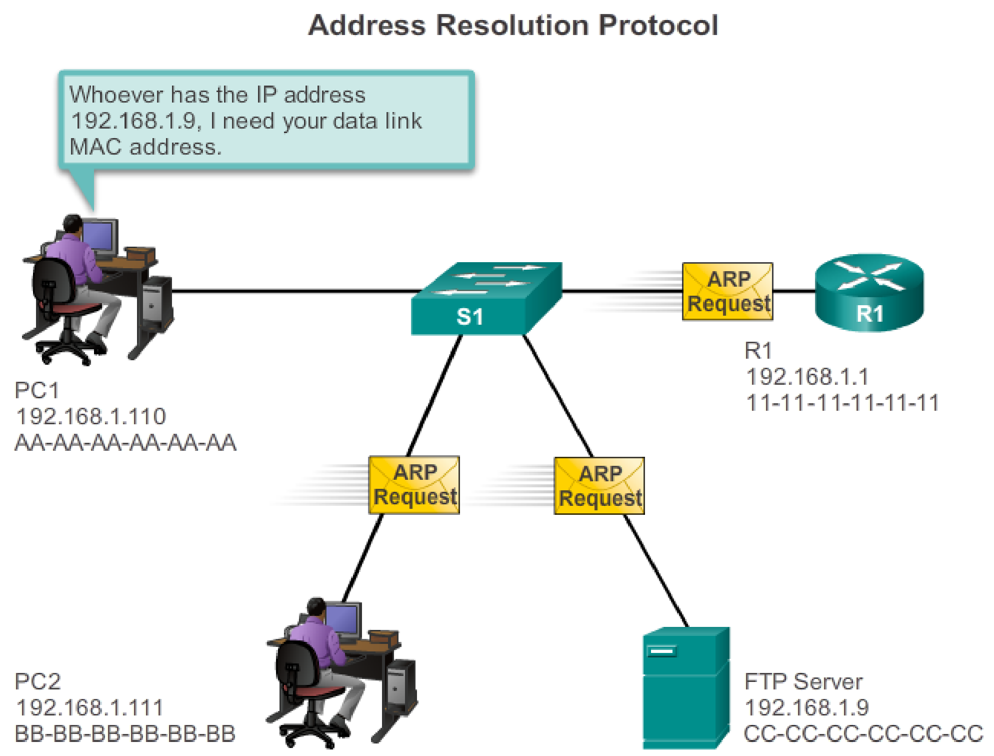
The ARP protocol provides **two basic functions:**

- Resolving IPv4 addresses to MAC addresses
- Maintaining a table of mappings – the ARP table



MAC and IP Addresses

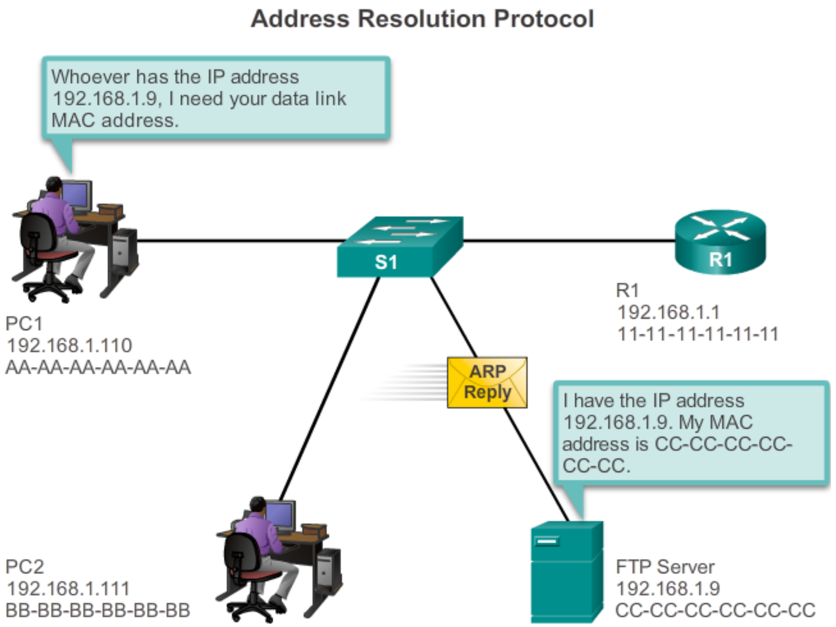
PC1 sends an ARP request to get MAC address of ftp server



MAC and IP Addresses

FTP server sends an ARP reply with his MAC address to PC1

PC1 can now assembly the frame



ARP TABLE

- shows the **relation between an ip address and a MAC address**
- stored in the RAM of the device
- each entry, or row, of the ARP table has a pair of values: an IP Address and a MAC address (data link layer address) = a map
  → you can locate an IP address in the table and discover the corresponding MAC address
- when a node receives frames from the media, it records the source IP and MAC address as a mapping in the ARP table
- to send a frame
  - the transmitting node attempts to find in the ARP table the MAC address mapped to the IPv4 destination
  - if the map is found, the node uses this MAC address as the destination MAC in the frame that encapsulates the IPv4 packet
  - the frame is then encoded onto the networking media

Maintaining the ARP table

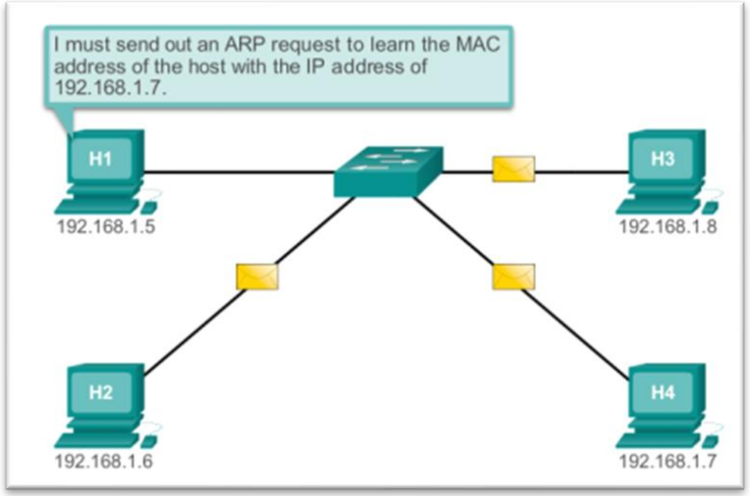
- The ARP table is maintained dynamically. There are two ways that a device can gather MAC addresses.
    - Monitor the traffic (source ip and source MAC)
    - ARP request (Destination MAC address: Layer 2 broadcast address / Broadcast MAC address: FF-FF-FF-FF-FF-FF)

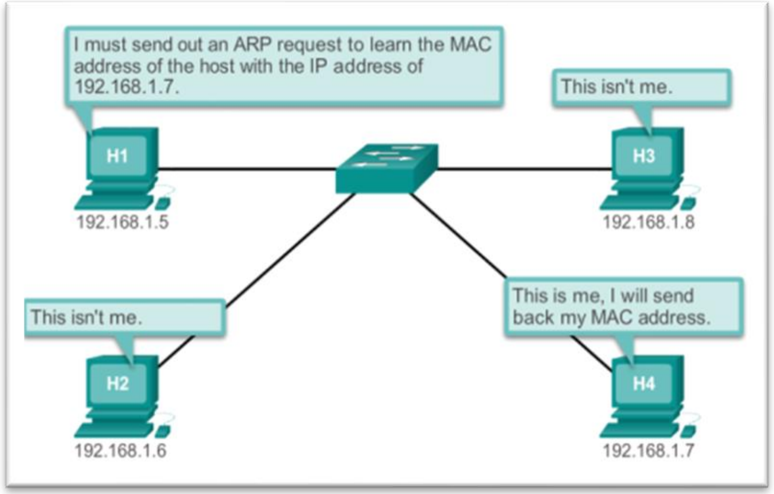
ARP request –

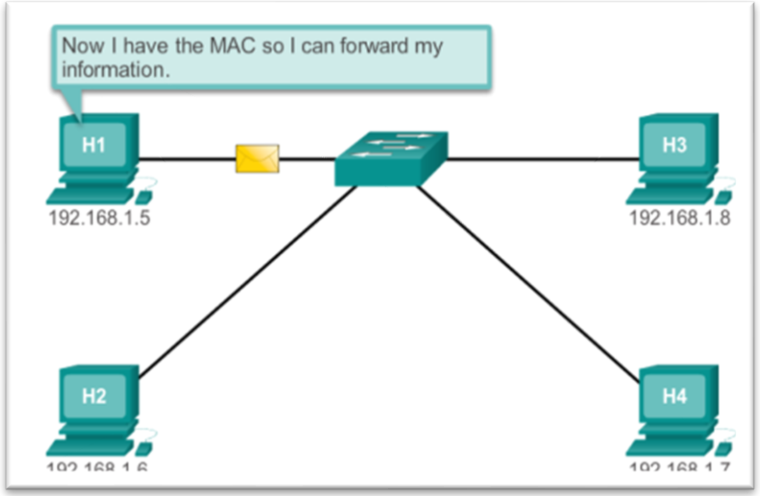
- Layer 2 broadcast to all devices on the Ethernet LAN
- The node that matches the IP address in the broadcast will reply (ARP reply)
- If no device responds to the ARP request, the packet is dropped because a frame cannot be created

Static map entries can be entered in an ARP table, but this is rarely done

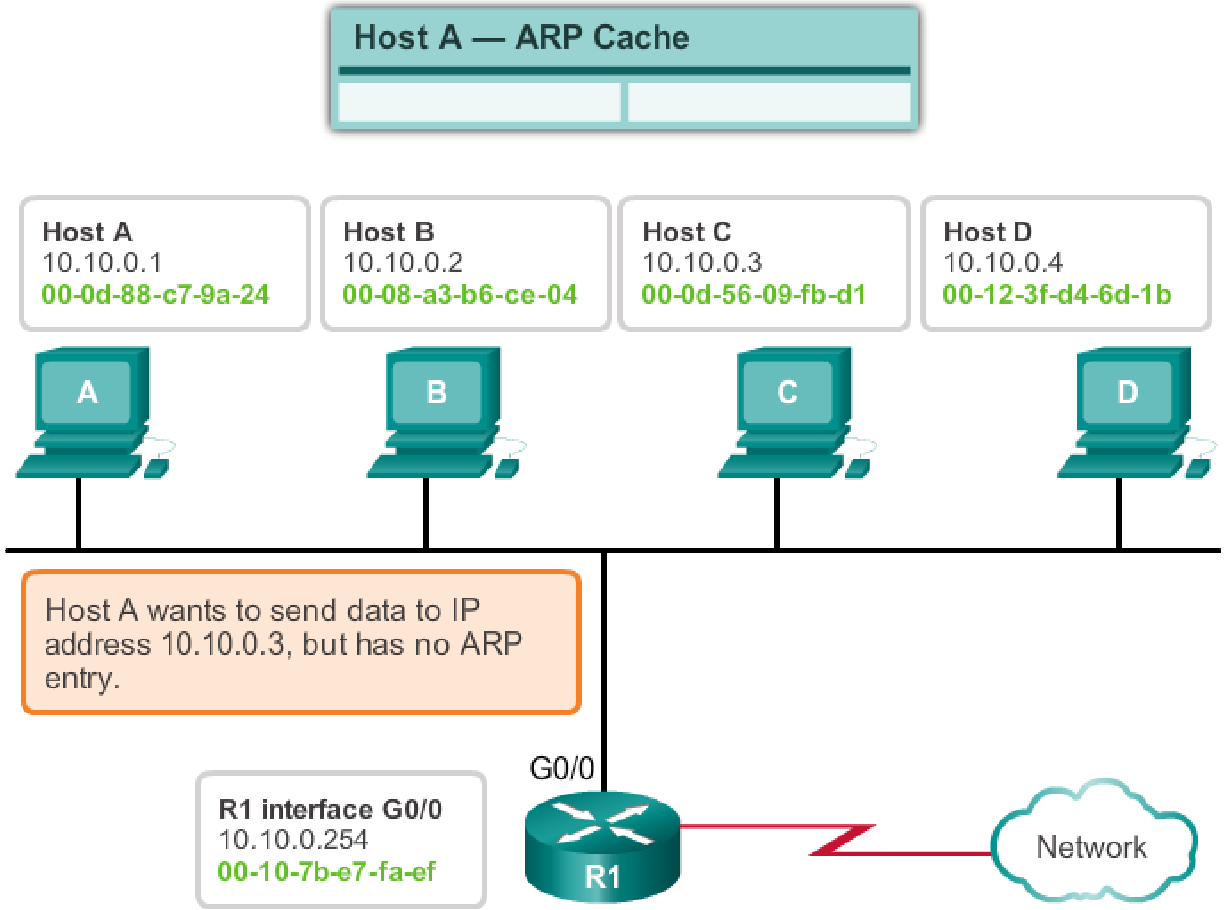
ARP operation



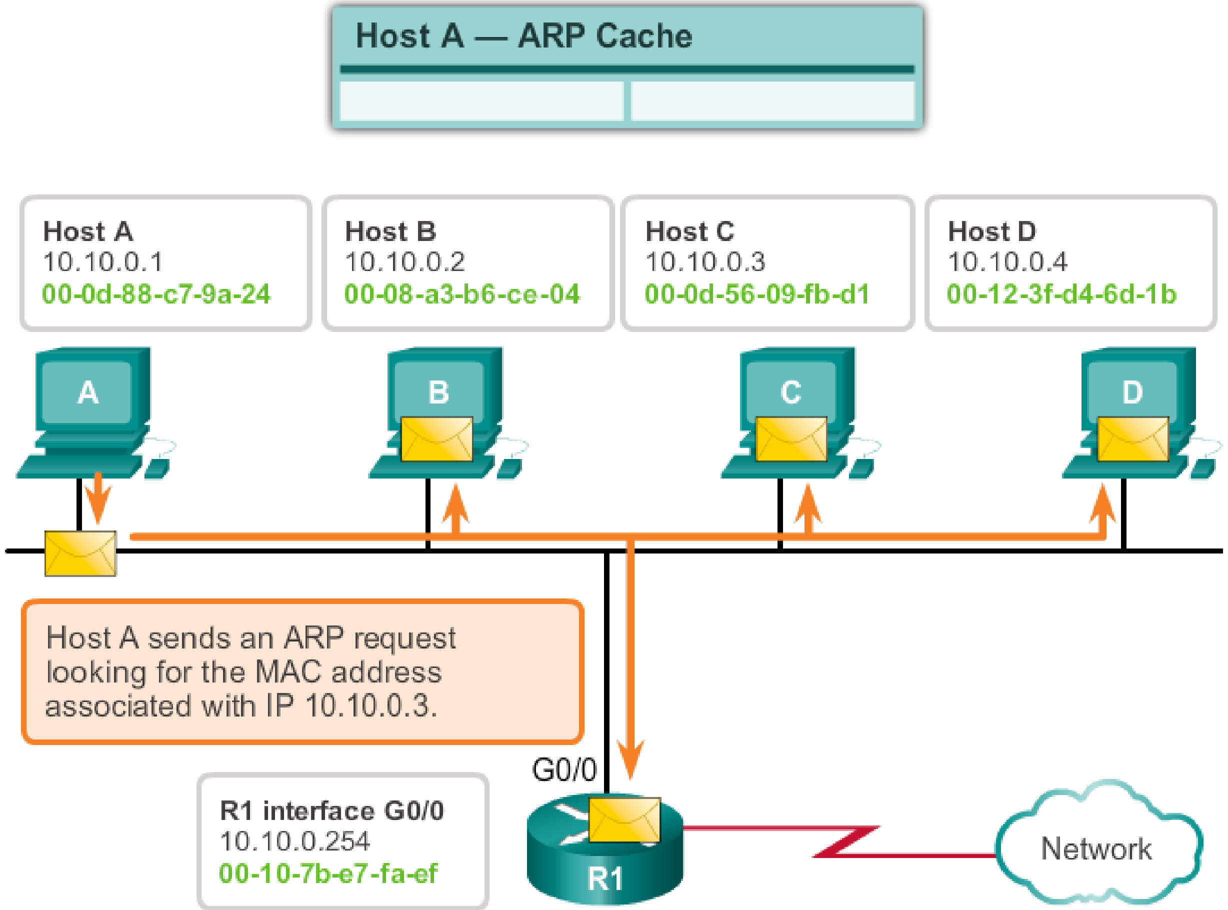




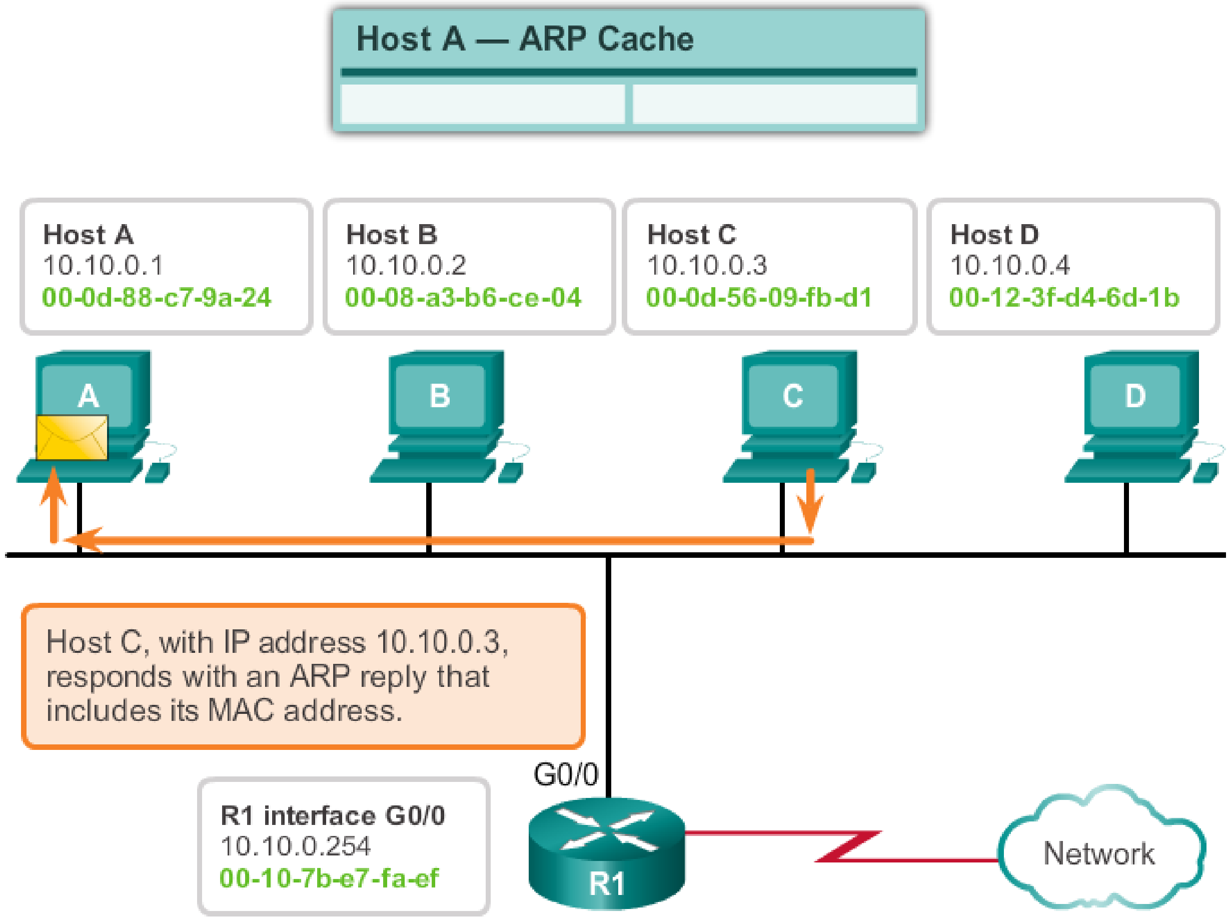
ARP operation



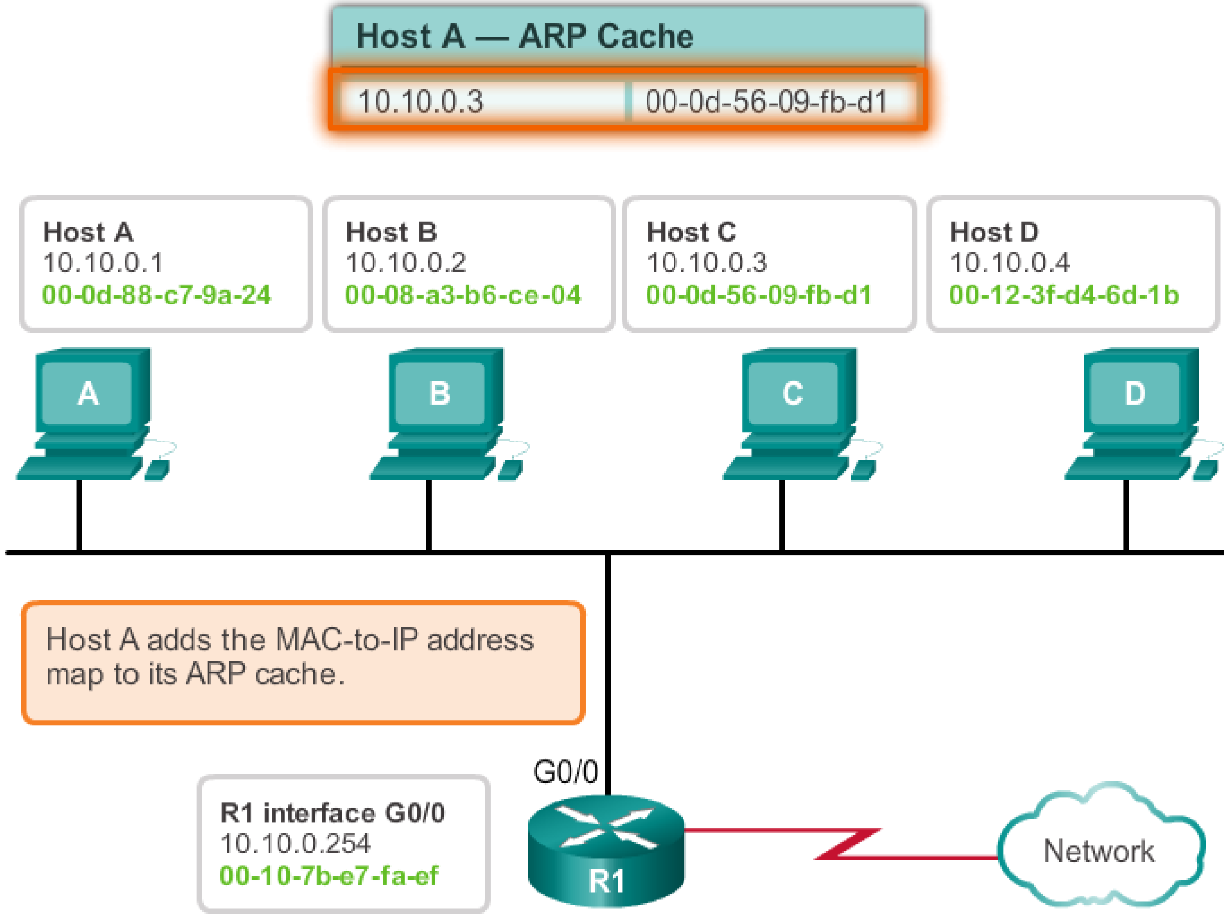
ARP operation



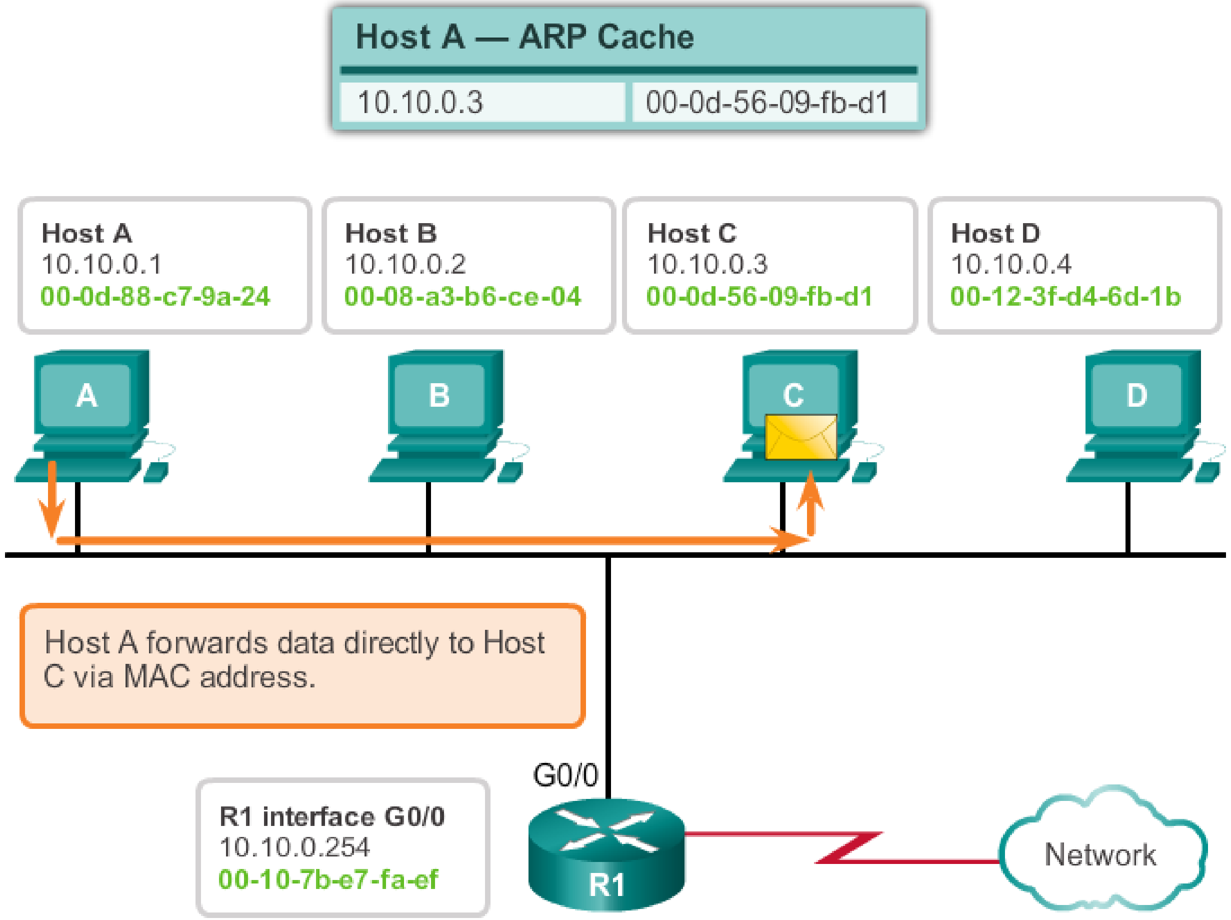
ARP operation



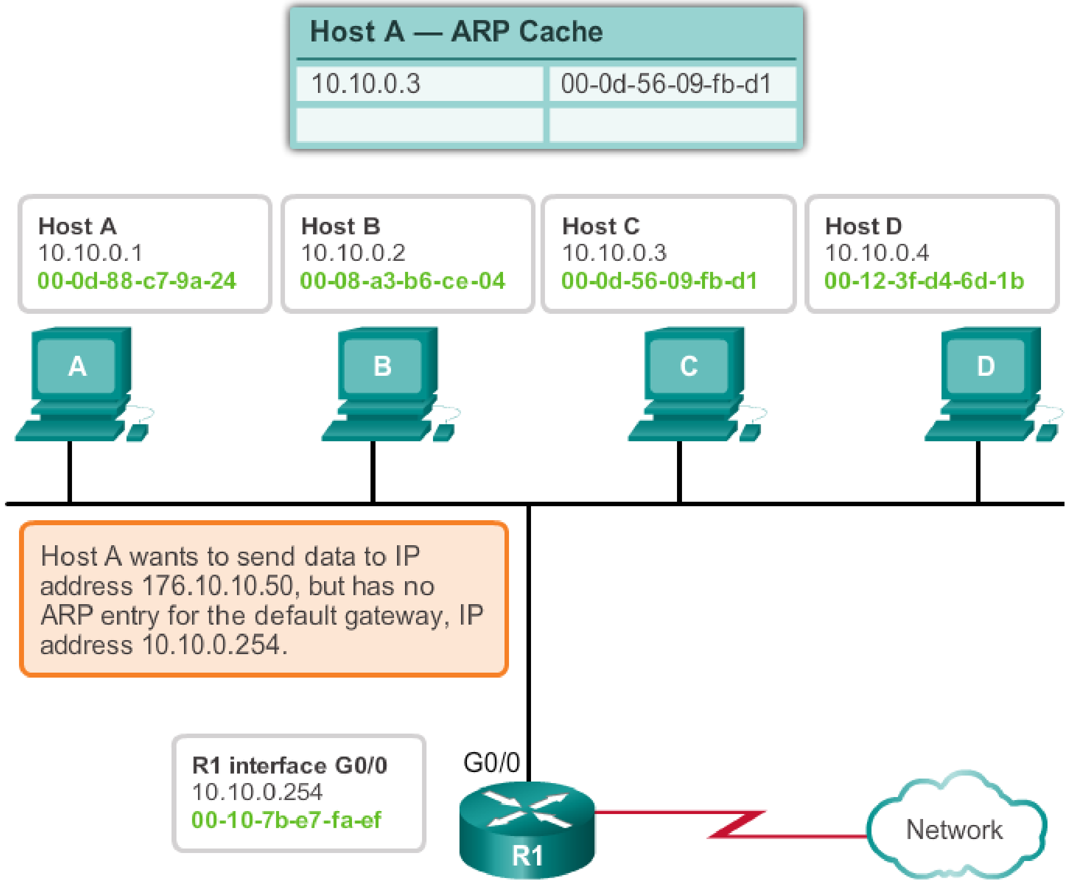
ARP operation



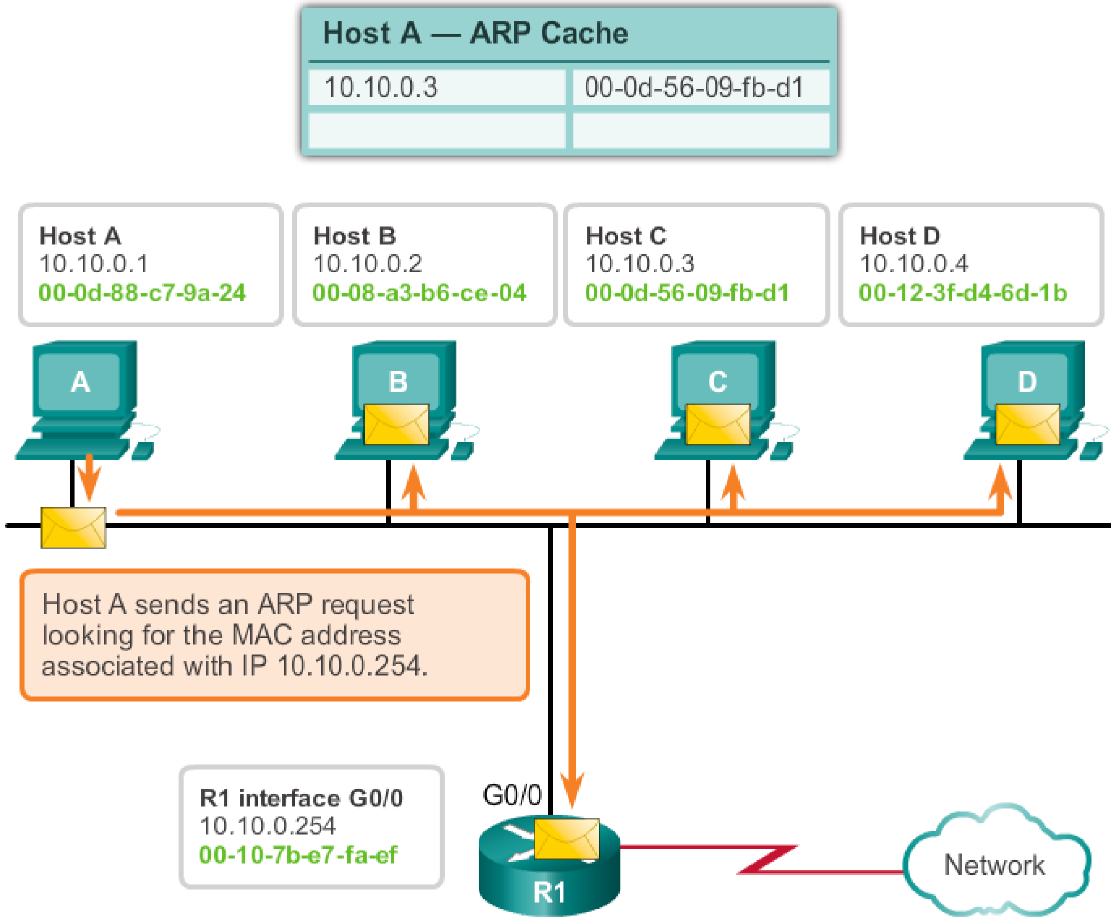
ARP operation



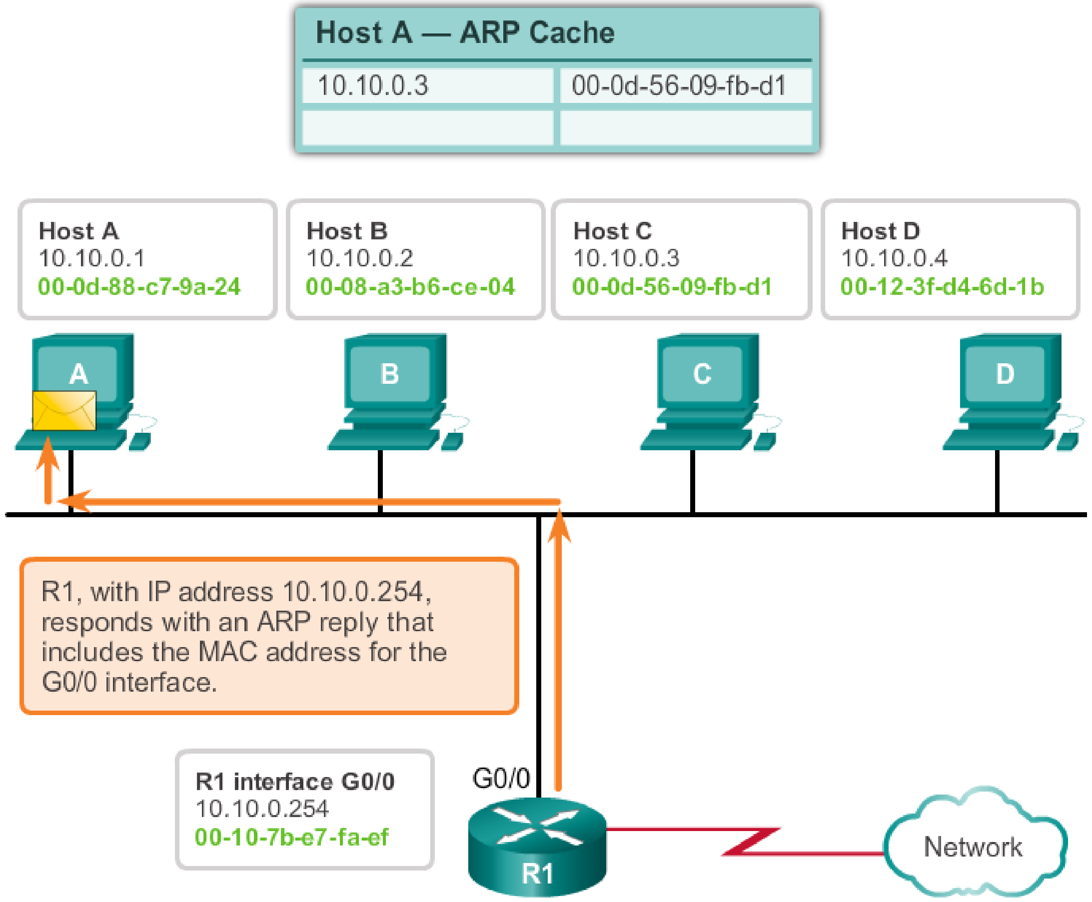
ARP operation



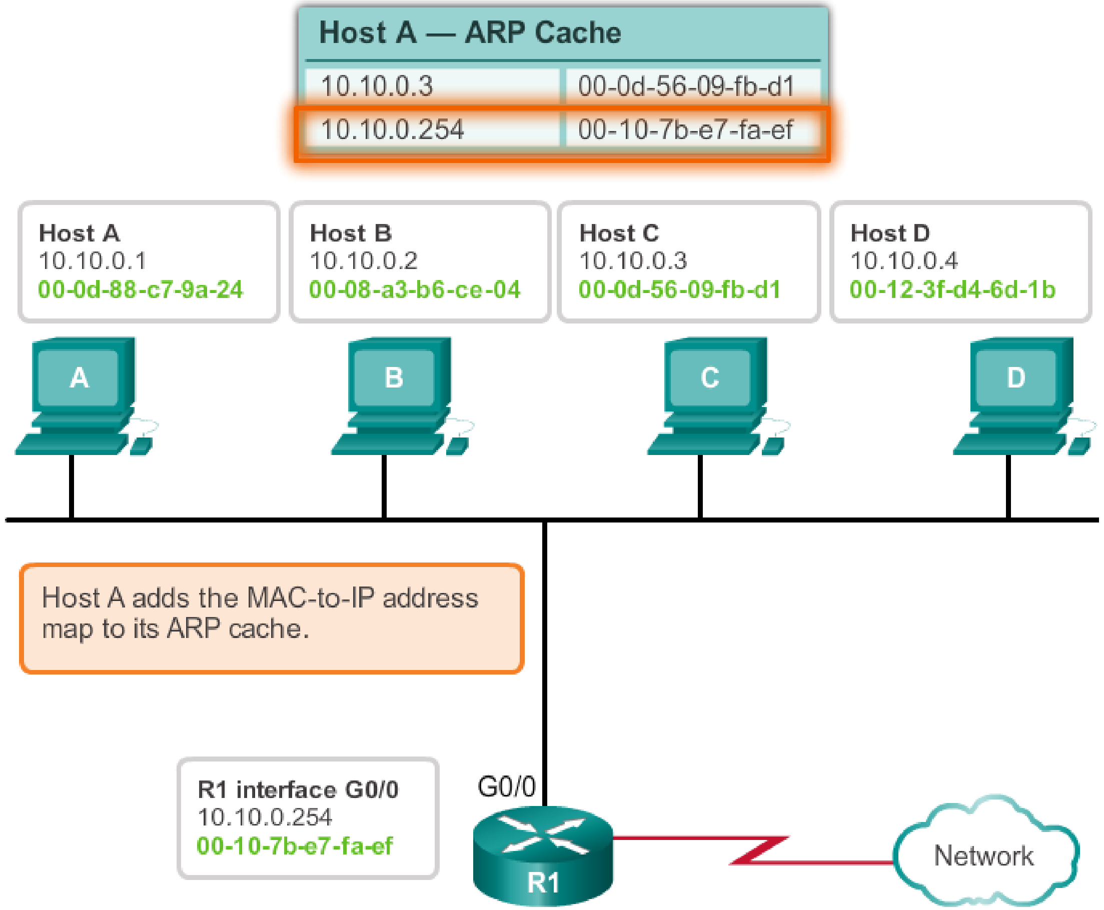
ARP operation



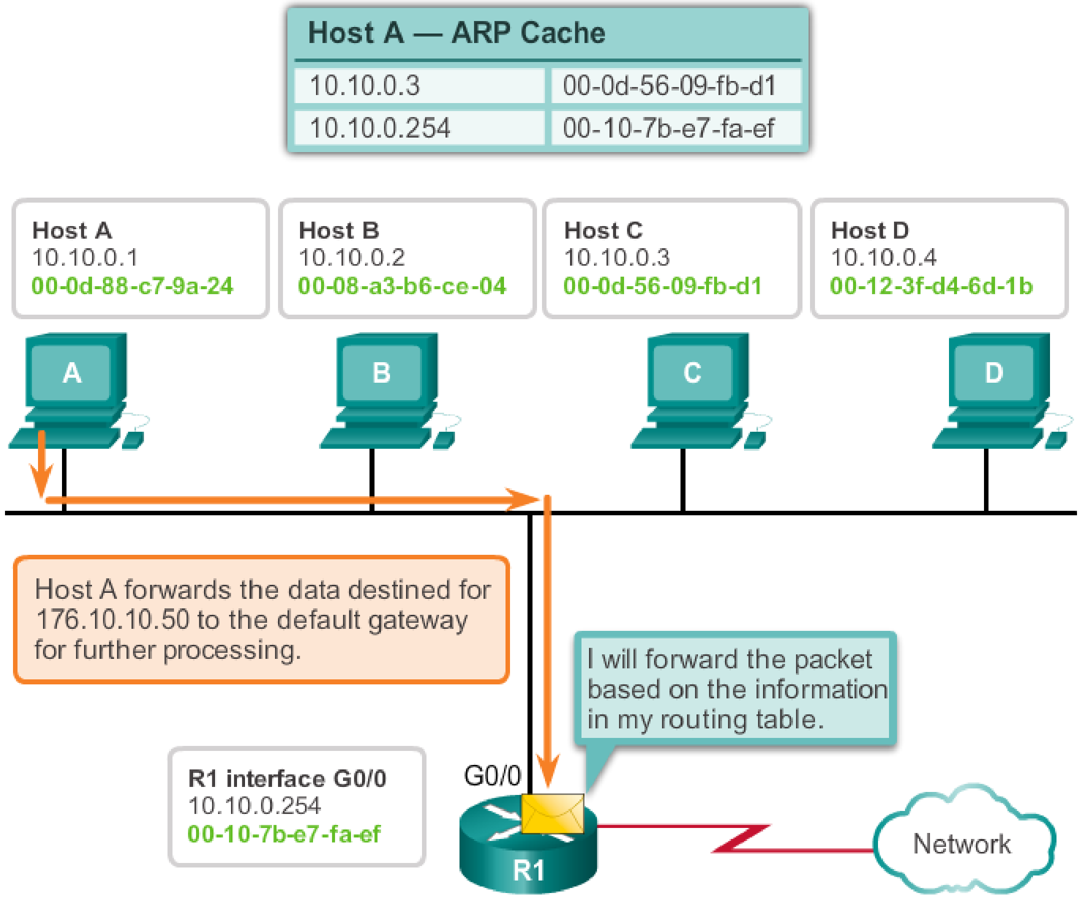
ARP operation



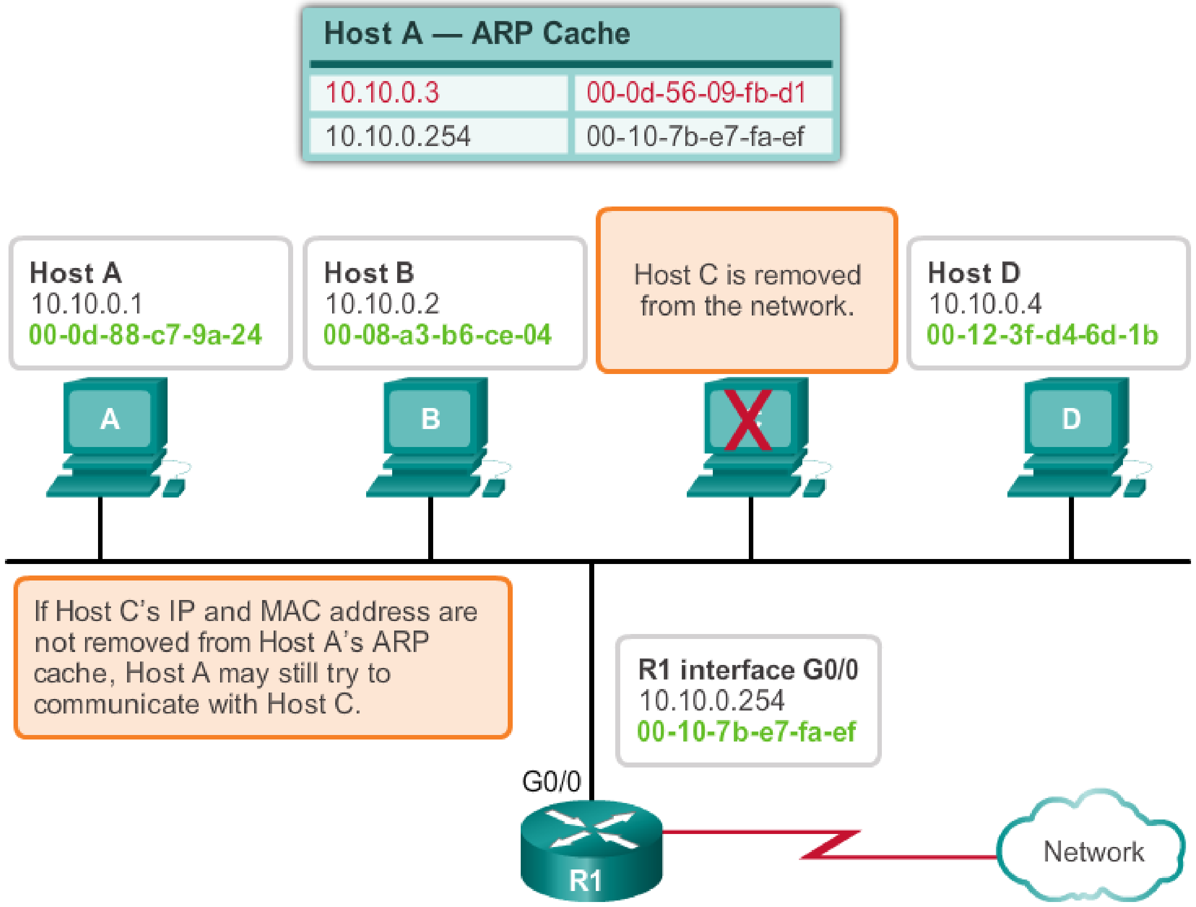
ARP operation



ARP operation



ARP operation



ARP Tables on Networking Devices

- On a Cisco router, the **show ip arp** command is used to display the ARP table

```
Router#show ip arp

                               Age
Protocol  Address          (min)  Hardware Addr   Type   Interface
Internet  172.16.233.229     -    0000.0c59.f892  ARPA   Ethernet0/0
Internet  172.16.233.218     -    0000.0c07.ac00  ARPA   Ethernet0/0
Internet  172.16.168.11      -    0000.0c63.1300  ARPA   Ethernet0/0
Internet  172.16.168.254     9    0000.0c36.6965  ARPA   Ethernet0/0
```

- On a Windows PC, the **arp –a** command is used to display the ARP table

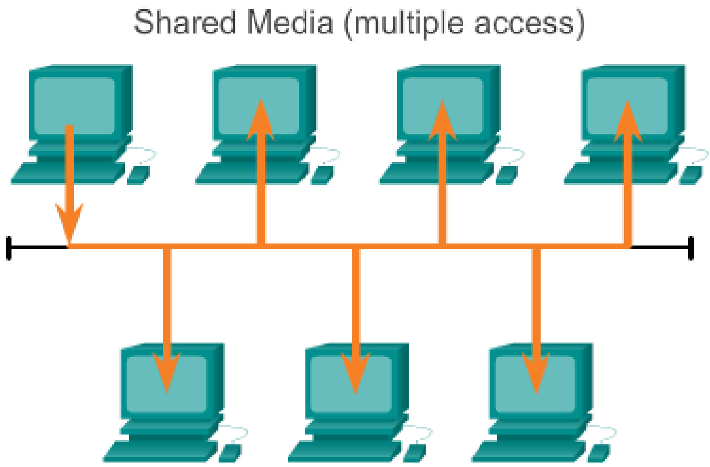
- **arp –d**: clear arp table

```
C:\>arp -a

Interface: 192.168.1.67 --- 0xa
  Internet Address      Physical Address      Type
  192.168.1.254         64-0f-29-0d-36-91     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

How ARP Can Create Problems

ARP broadcasts can flood the local media.



ARP Issues:

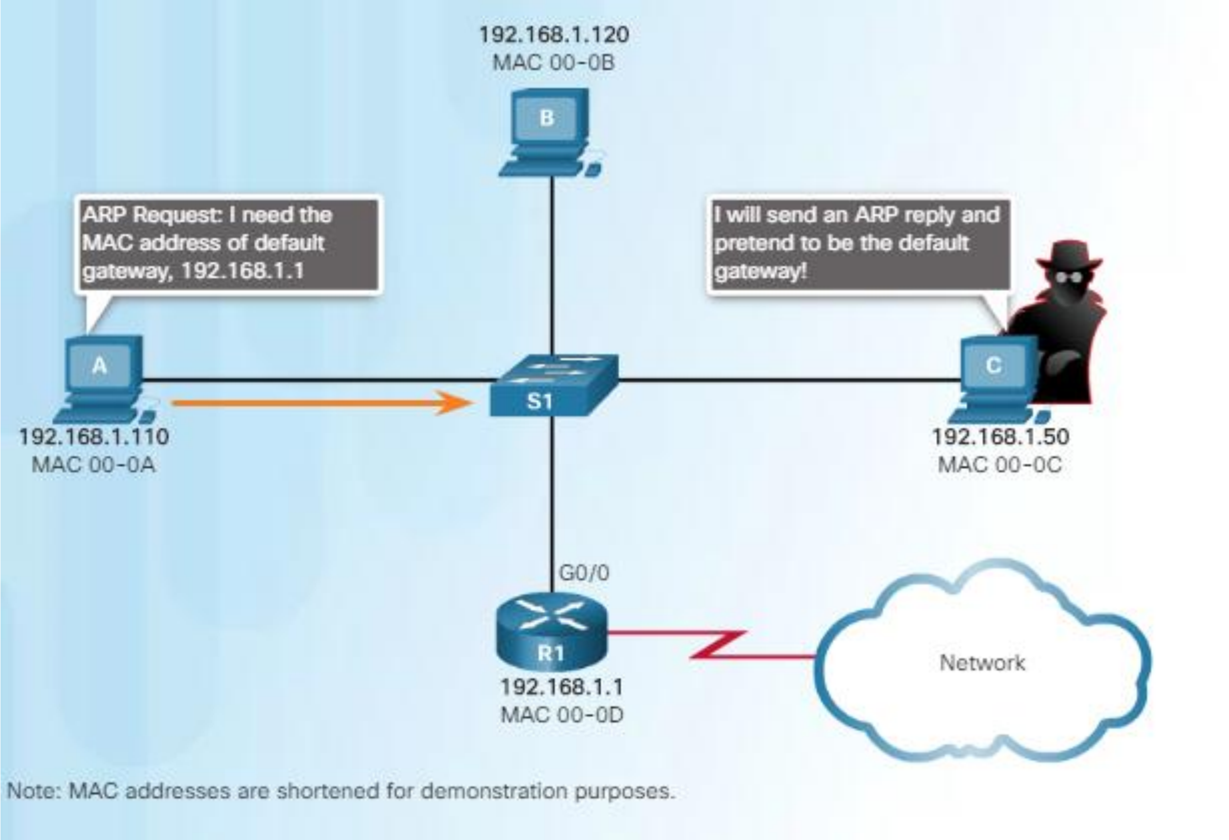
- Broadcasts, overhead on the media
- Security

A false ARP message can provide an incorrect MAC address that will then hijack frames using that address (called a spoof).

ARP Spoofing/ARP table poisoning

- Sending ARP responses for another device
- If the spoofers response comes in before the actual response, the wrong MAC address is put into the ARP table

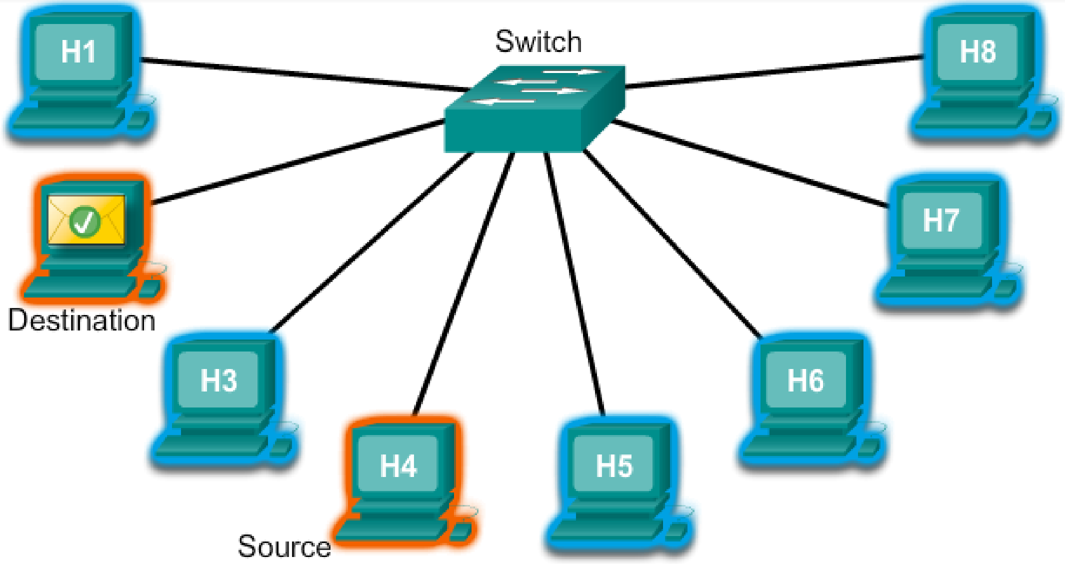
Man-in-the-middle attack



Switch Port Fundamentals

MAC Table

fa0/1	fa0/2	fa0/3	fa0/4
206d.8c01.0000	206d.8c01.1111	206d.8c01.2222	206d.8c01.3333
fa0/5	**fa0/6**	**fa0/7**	**fa0/8**
206d.8c01.4444	206d.8c01.5555	206d.8c01.6666	206d.8c01.7777

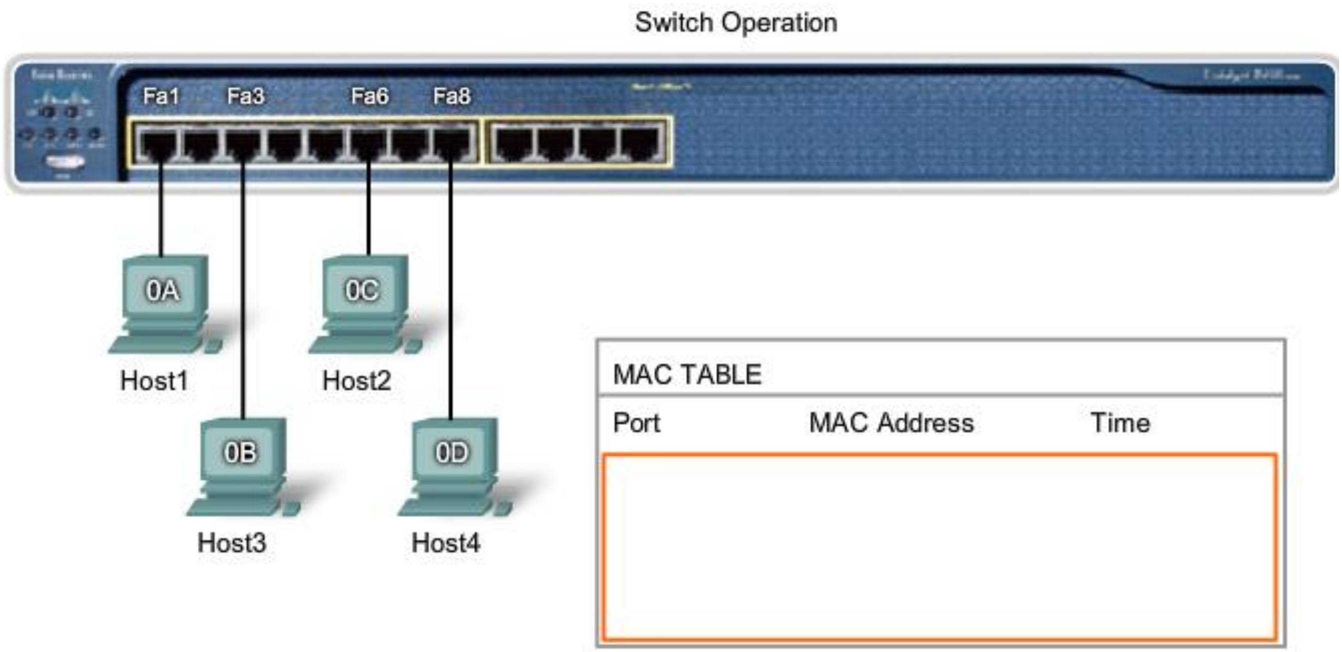


Switch Port Fundamentals

- Ethernet Switches → Selective forwarding
  - When a switch knows on which port the destination MAC is, it will send the data only on this port (= selective forwarding)
  - Looks like a **temporary point to point connection between both endpoints** in which the connection is limited to sending a single frame

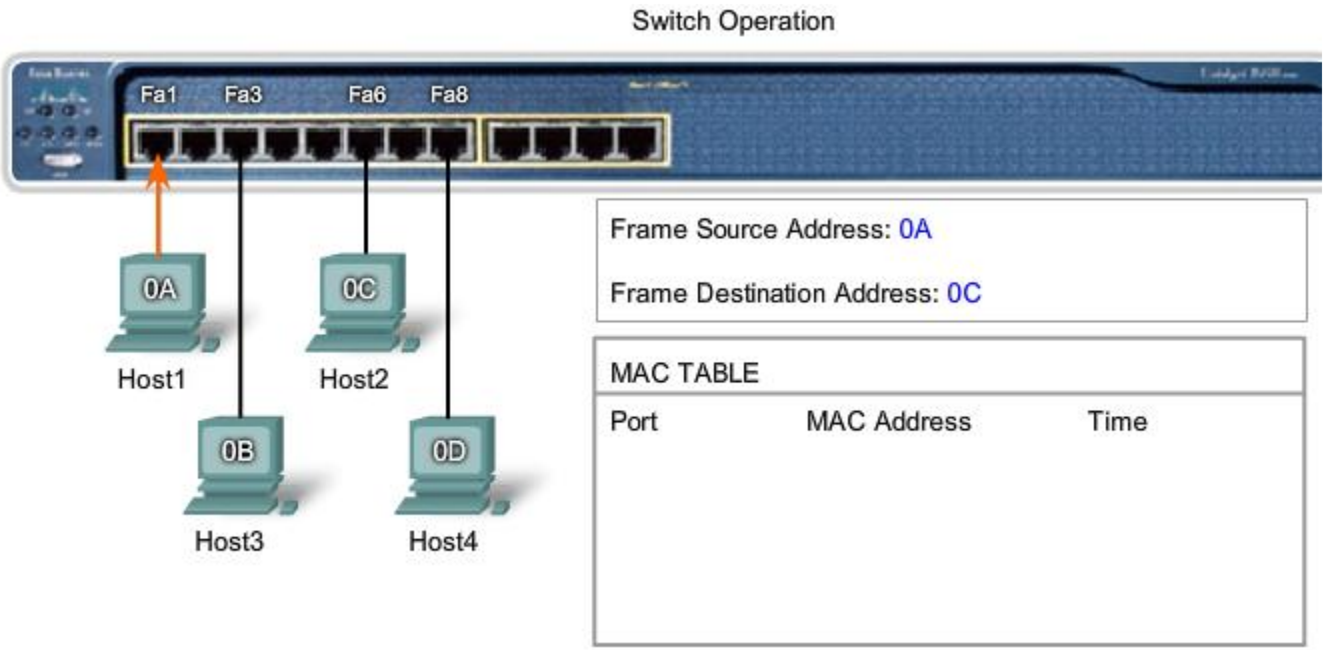
  - **Learning**
    - As a new frame enters the switch, it examines the **source MAC** address.
- **Aging**
  - The entries in the MAC table are **time stamped**. After a specific time, the entry is deleted
- **Flooding** send the frame to anyone except its own
- **Selective Forwarding**
  - the process of examining a frame's destination MAC address and forwarding it to the appropriate port
- **Filtering**
  - In some cases, a frame is not forwarded = frame filtering

Switch Port Fundamentals



Upon initialization of the switch, the MAC address table is empty.

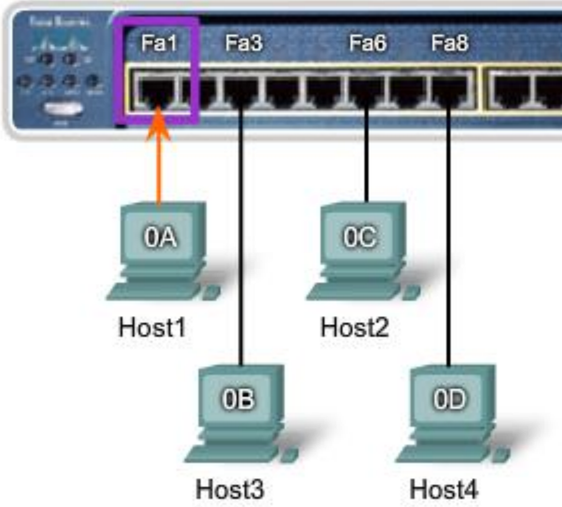
Switch Port Fundamentals



Host1 sends data to Host2. The frame sent contains both a source MAC address and a destination MAC address.

Switch Port Fundamentals

Switch Operation



Frame Source Address: 0A

Frame Destination Address: 0C

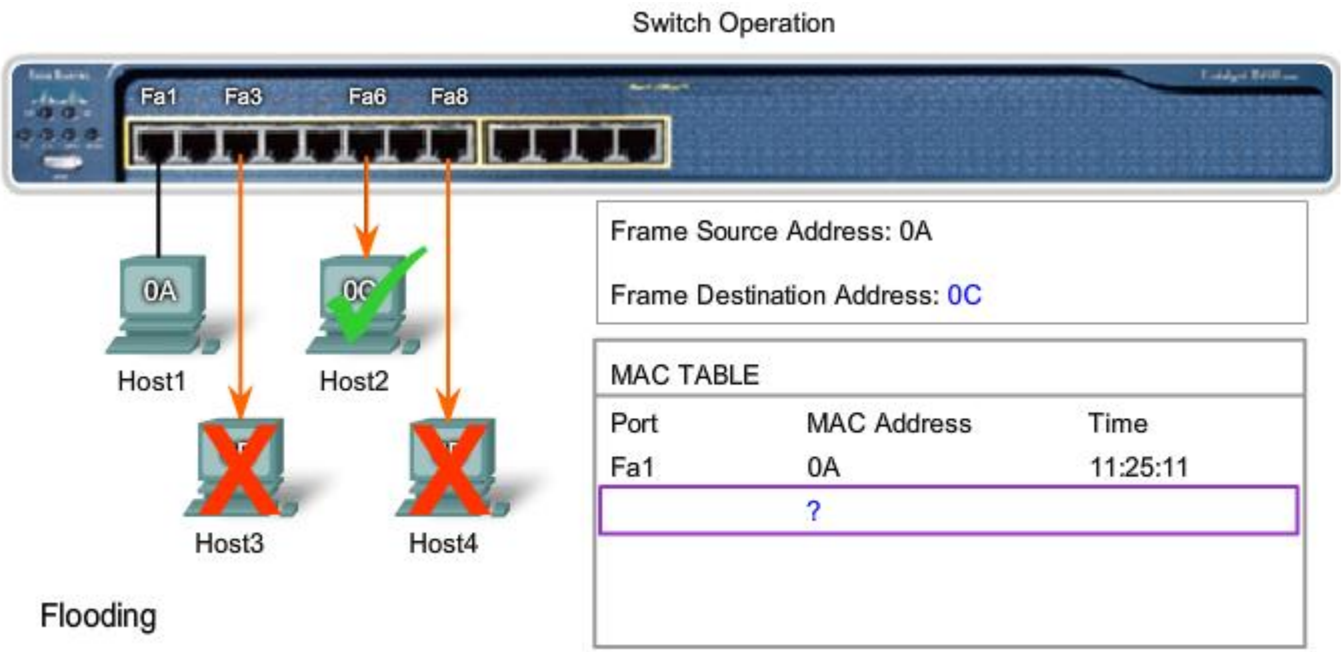
MAC TABLE

Port	MAC Address	Time
Fa1	0A	11:25:11

Learning

The switch reads the source MAC address, 0A, from the frame received on port Fa1 and stores it in the MAC address table for use in the forwarding of frames to Host1.

Switch Port Fundamentals

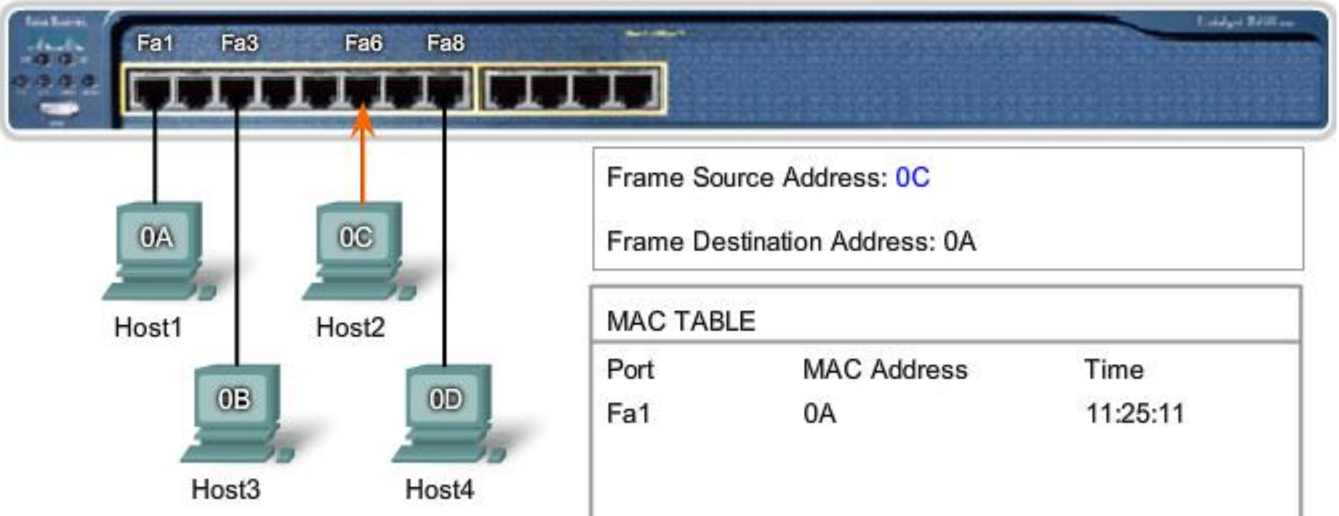


Flooding

The destination MAC address, 0C, is not in the MAC Table. The switch floods the frame out all ports except port Fa1, the port for the sender. Host3 and Host4 receive the frame, but the address in the frame does not match their MAC address. They drop the frame. The destination MAC address in the frame matches Host2 and it accepts the frame.

Switch Port Fundamentals

Switch Operation



Frame Source Address: 0C

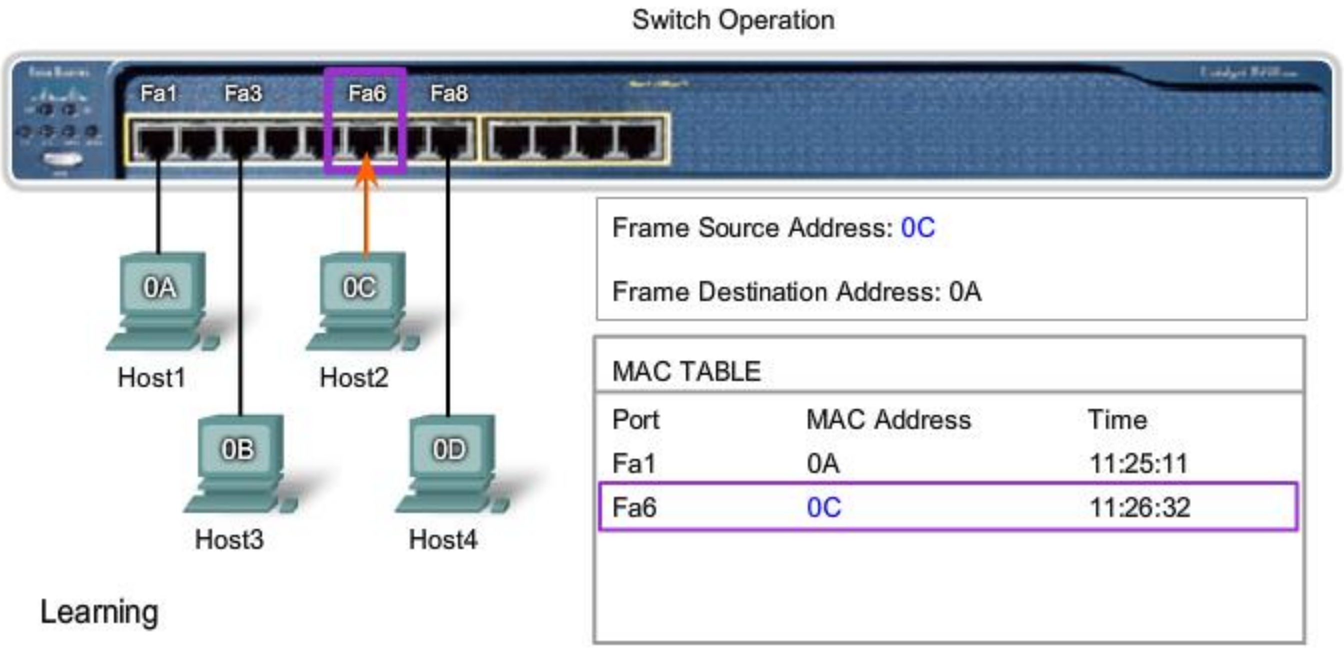
Frame Destination Address: 0A

MAC TABLE

Port	MAC Address	Time
Fa1	0A	11:25:11

Host2 sends a frame to Host1 containing a reply. The source address in the frame is the MAC address of Host2. The destination address in the frame matches the MAC address for Host1.

Switch Port Fundamentals

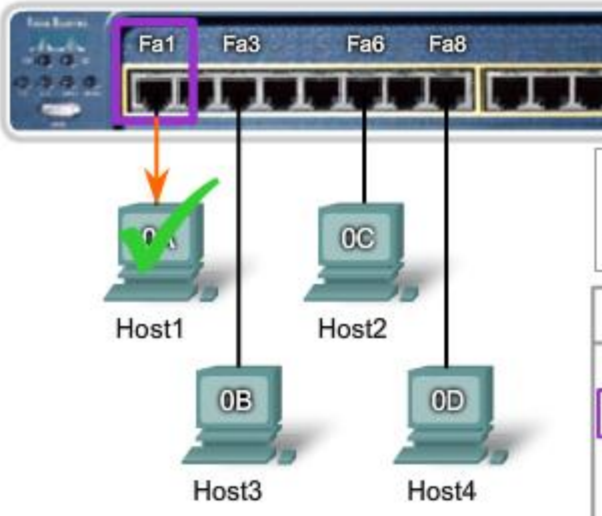


Learning

The switch reads the source MAC address, 0C, from the frame received on port Fa6, and stores it in the MAC address table for use in the forwarding of frames to Host2.

Switch Port Fundamentals

Switch Operation



Frame Source Address: 0C

Frame Destination Address: 0A

MAC TABLE

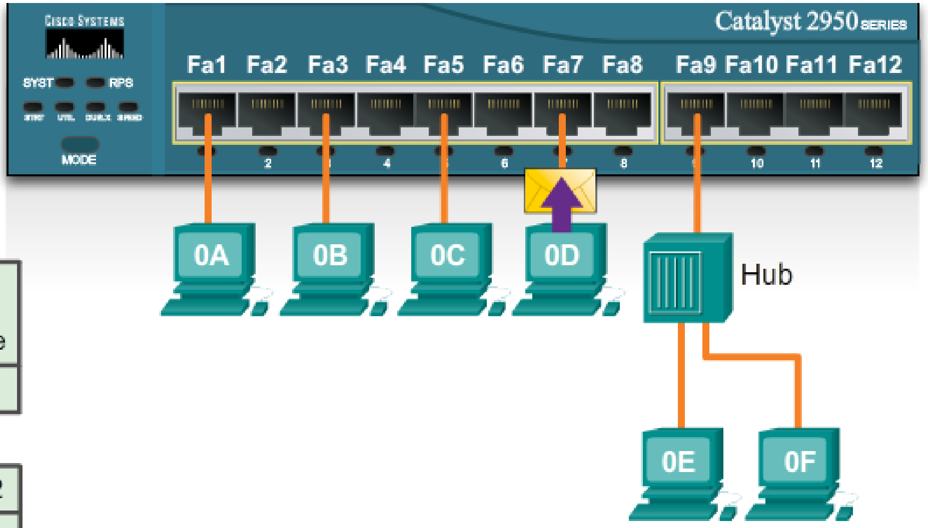
Port	MAC Address	Time
Fa1	0A	11:25:11
Fa6	0C	11:26:32

Selective Forwarding

The destination MAC address, 0A, is in the MAC address table. The switch selectively forwards the frame out port Fa1 only. The destination MAC address in the frame matches the MAC address for Host1. Host1 accepts the frame.

Switch Port Fundamentals

Activity
Determine how the switch forwards a frame based on the Source MAC and Destination MAC addresses and information in the switch MAC table.
Answer the questions below using the information provided.



Frame

Preamble	Destination MAC	Source MAC	Length Type	Encapsulated Data	End of Frame
	0F	0D			

MAC Table

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
		0B									

Question 1 - Where will the switch forward the frame?

☐ Fa1 ☐ Fa2 ☐ Fa3 ☐ Fa4 ☐ Fa5 ☐ Fa6 ☐ Fa7 ☐ Fa8 ☐ Fa9 ☐ Fa10 ☐ Fa11 ☐ Fa12

Question 2 - When the switch forwards the frame, which statement(s) are true?

- ☐ Switch adds the source MAC address to the MAC table.
- ☐ Frame is a broadcast frame and will be forwarded to all ports.
- ☐ Frame is a unicast frame and will be sent to specific port only.
- ☐ Frame is a unicast frame and will be flooded to all ports.
- ☐ Frame is a unicast frame but it will be dropped at the switch.

Check

New Problem

Help

Switch Port Fundamentals

Activity
Determine how the switch forwards a frame based on the Source MAC and Destination MAC addresses and information in the switch MAC table.
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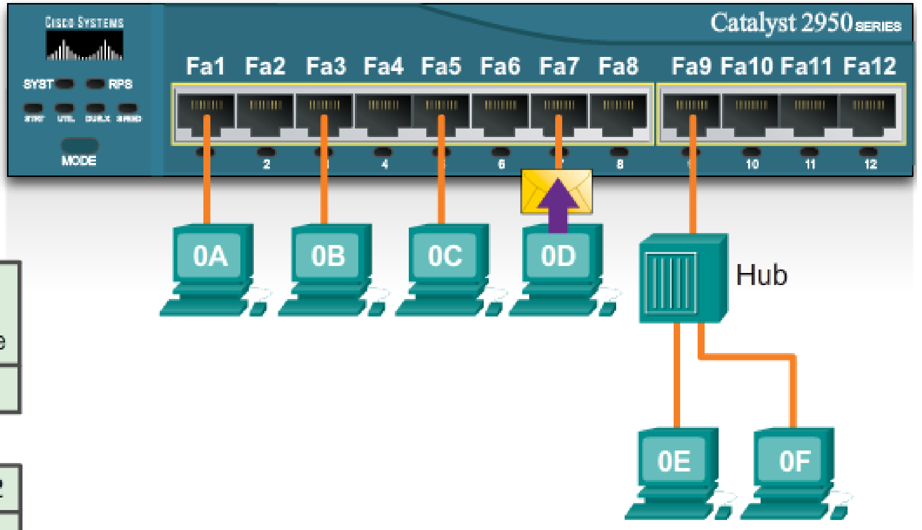

Frame

Preamble	Destination MAC	Source MAC	Length Type	Encapsulated Data	End of Frame
	0F	**0D**			

MAC Table

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
		0B									

Question 1 - Where will the switch forward the frame?

[x] Fa1 [ ] Fa2 [x] Fa3 [ ] Fa4 [x] Fa5 [ ] Fa6 [ ] Fa7 [ ] Fa8 [x] Fa9 [ ] Fa10 [ ] Fa11 [ ] Fa12

Question 2 - When the switch forwards the frame, which statement(s) are true?

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- [ ] Frame is a unicast frame and will be sent to specific port only.
- [x] Frame is a unicast frame and will be flooded to all ports.
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Check

New Problem

Help

Switch Port Fundamentals

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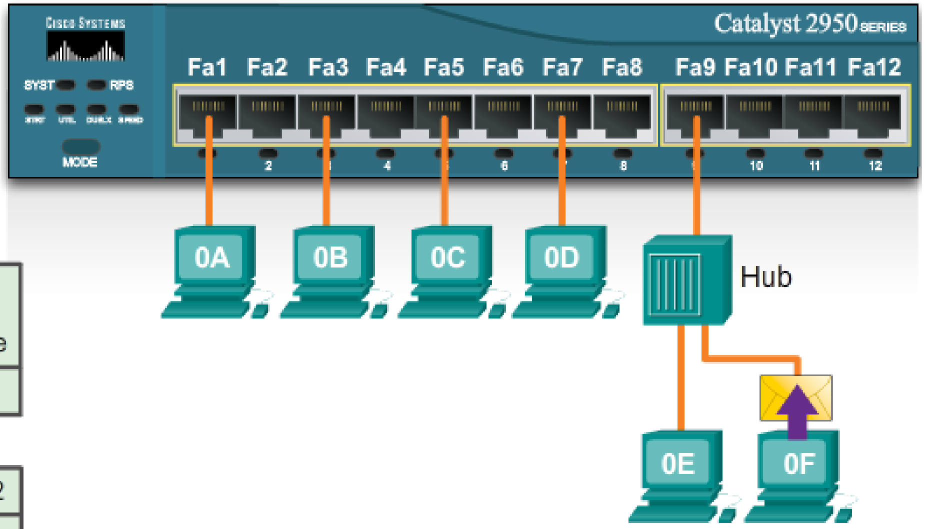

Frame

Preamble	Destination MAC	Source MAC	Length Type	Encapsulated Data	End of Frame
	0A	0F			

MAC Table

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
0A				0C				0F			

Question 1 - Where will the switch forward the frame?

☐ Fa1 ☐ Fa2 ☐ Fa3 ☐ Fa4 ☐ Fa5 ☐ Fa6 ☐ Fa7 ☐ Fa8 ☐ Fa9 ☐ Fa10 ☐ Fa11 ☐ Fa12

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New Problem

Help

Switch Port Fundamentals

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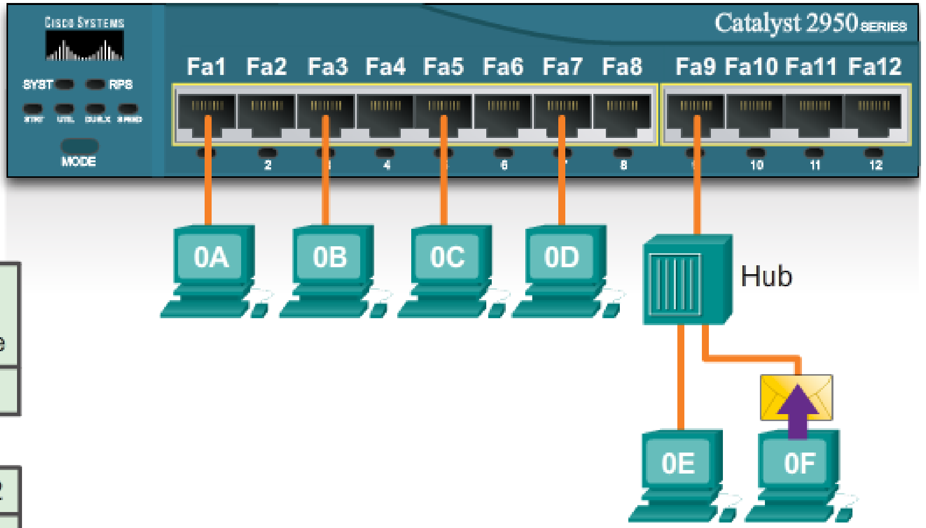

Frame

Preamble	Destination MAC	Source MAC	Length Type	Encapsulated Data	End of Frame
	0A	0F			

MAC Table

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
0A				0C				0F			

Question 1 - Where will the switch forward the frame?

- [x] Fa1
- [ ] Fa2
- [ ] Fa3
- [ ] Fa4
- [ ] Fa5
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- [ ] Fa9
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Check

New Problem

Help

Switch Port Fundamentals

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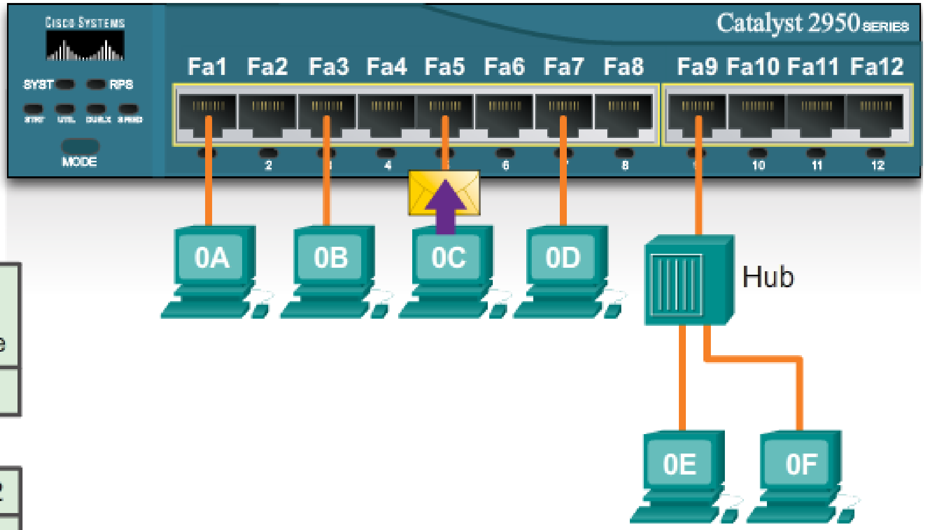

Frame

Preamble	Destination MAC	Source MAC	Length Type	Encapsulated Data	End of Frame
	FF	**0C**			

MAC Table

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
0A		**0B**		**0C**		**0D**		**0F**			

Question 1 - Where will the switch forward the frame?

- [ ] Fa1 - [ ] Fa2 - [ ] Fa3 - [ ] Fa4 - [ ] Fa5 - [ ] Fa6 - [ ] Fa7 - [ ] Fa8 - [ ] Fa9 - [ ] Fa10 - [ ] Fa11 - [ ] Fa12

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Check

New Problem

Help

Switch Port Fundamentals

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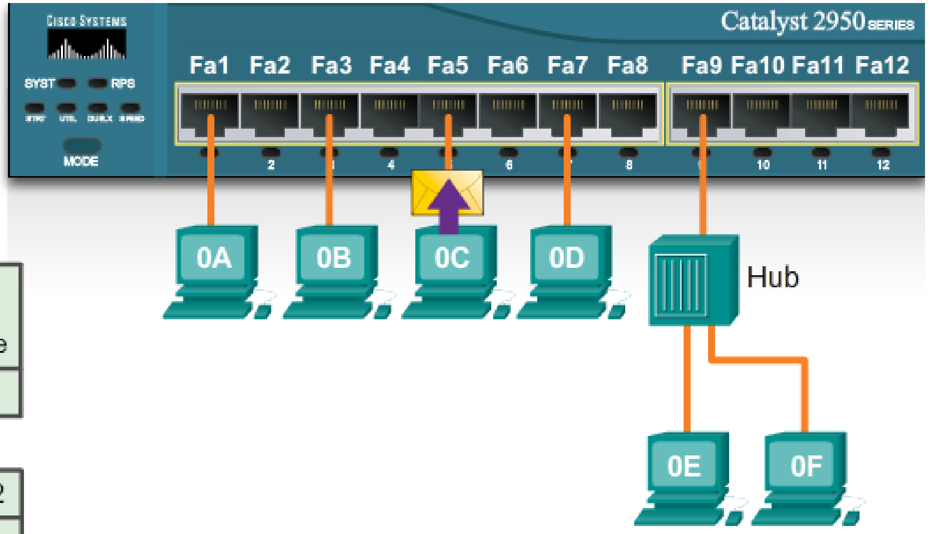

Frame

Preamble	Destination MAC	Source MAC	Length Type	Encapsulated Data	End of Frame
	FF	**0C**			

MAC Table

Fa1	Fa2	Fa3	Fa4	Fa5	Fa6	Fa7	Fa8	Fa9	Fa10	Fa11	Fa12
0A		**0B**		**0C**		**0D**		**0F**			

Question 1 - Where will the switch forward the frame?

- [x] Fa1
- [ ] Fa2
- [x] Fa3
- [ ] Fa4
- [ ] Fa5
- [ ] Fa6
- [x] Fa7
- [ ] Fa8
- [x] Fa9
- [ ] Fa10
- [ ] Fa11
- [ ] Fa12

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- [ ] Frame is a unicast frame and will be flooded to all ports.
- [ ] Frame is a unicast frame but it will be dropped at the switch.

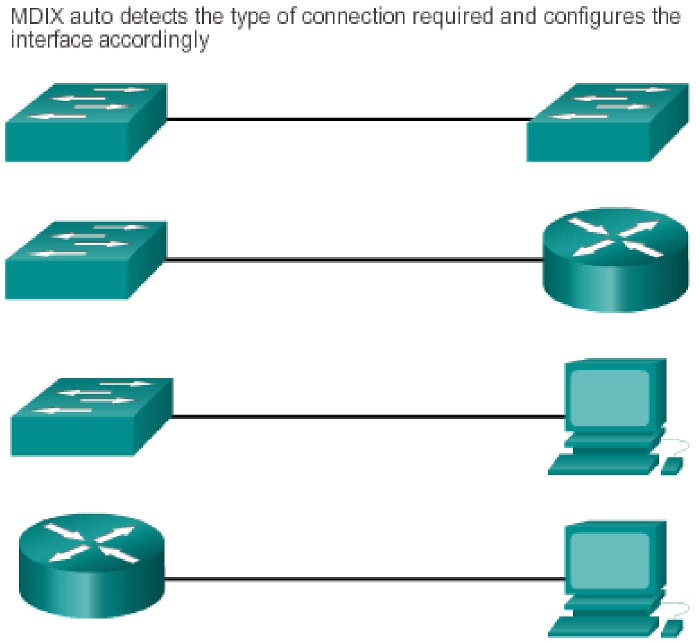
Check

New Problem

Help

Auto-MDIX

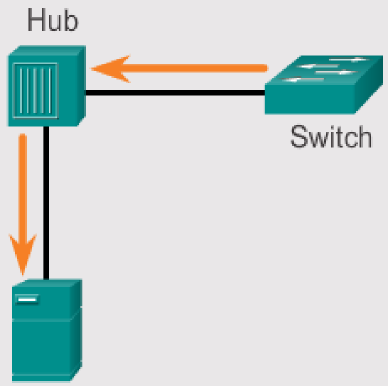
- The correct cable type needs to be defined for each port
- Most switch devices now support the **mdix auto** interface configuration command to enable the automatic medium-dependent interface crossover (auto-MDIX) feature
- When the auto-MDIX feature is enabled → the switch detects the required cable type for copper Ethernet connections and configures the interfaces accordingly



Duplex Settings

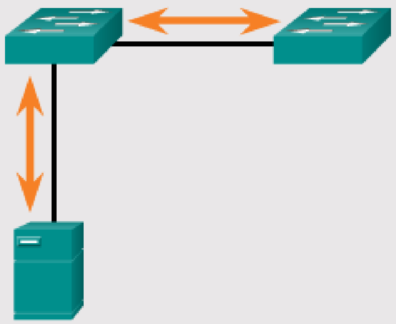
Half Duplex (CSMA/CD)

- Unidirectional data flow
- Higher potential for collision
- Hub connectivity

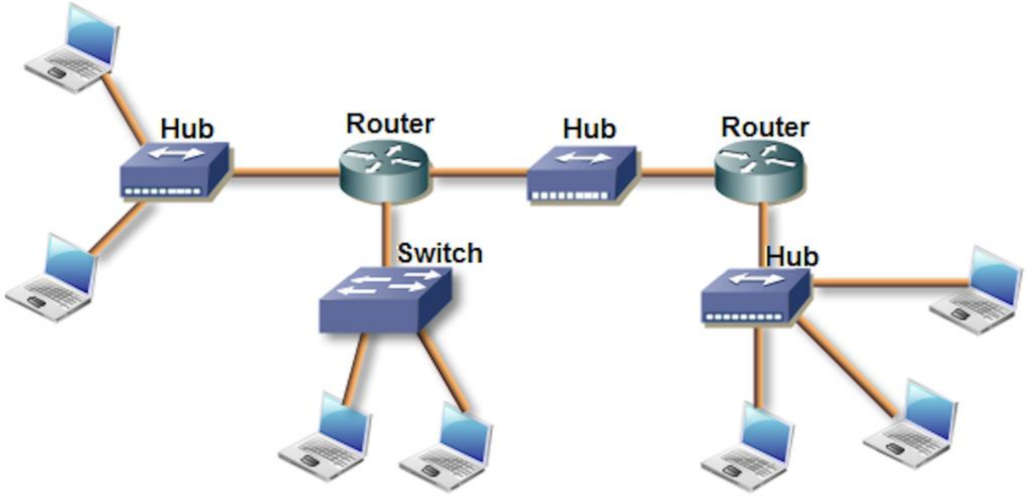


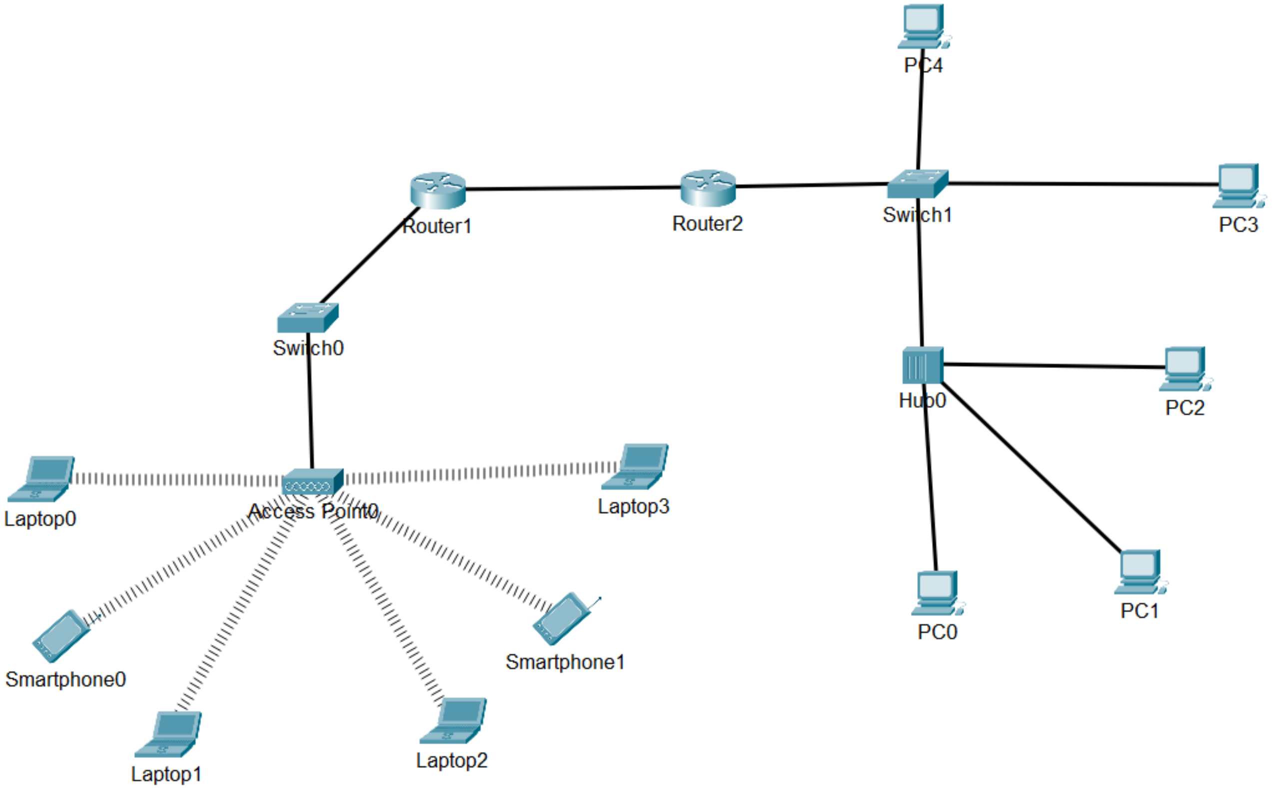
Full Duplex

- Point-to-point only
- Attached to dedicated switched port
- Requires full-duplex support on both ends
- Collision-free
- Collision detect circuit disabled



Collision domain vs broadcast domain





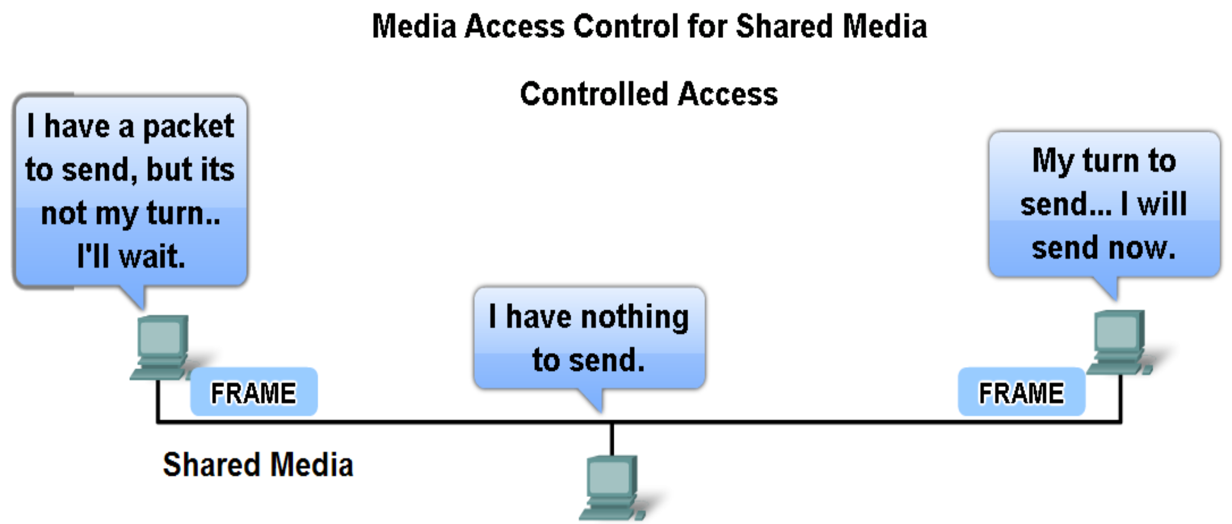
Media Access Control Techniques

Some network topologies share a common medium with different nodes
There are two basic media access control methods for shared media:

- **Controlled Access** (token ring, FDDI)
- **Contention-based Access** (ethernet)

Otherwise too much collisions

Controlled Access



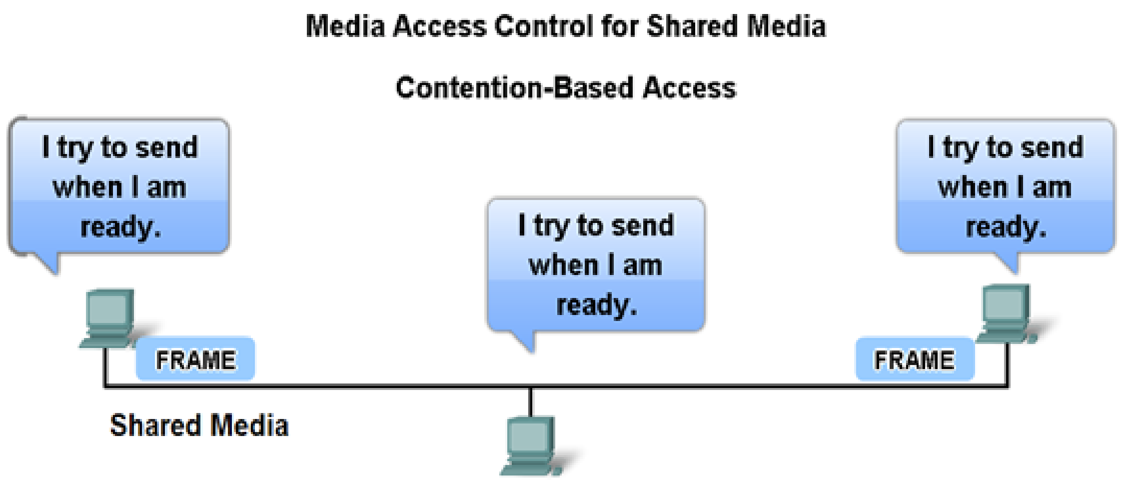
Method	Characteristics	Example
Controlled Access	• Only one station transmits at a time • Devices wishing to transmit must wait their turn • No collisions • Some deterministic networks use token passing	• Token Ring • FDDI

Contention-Based Access

- Allow any device that has data to send to try to access the medium
- To prevent complete chaos on the media, these methods use **a Carrier Sense Multiple Access (CSMA)** process to first detect if the media is carrying a signal

Carrier signal on the media from another node?

Yes:
Another device is transmitting
The media is busy
This device will wait and
tries again after a short time period
No:
The media is free
The device transmits its data



Contention-Based Access

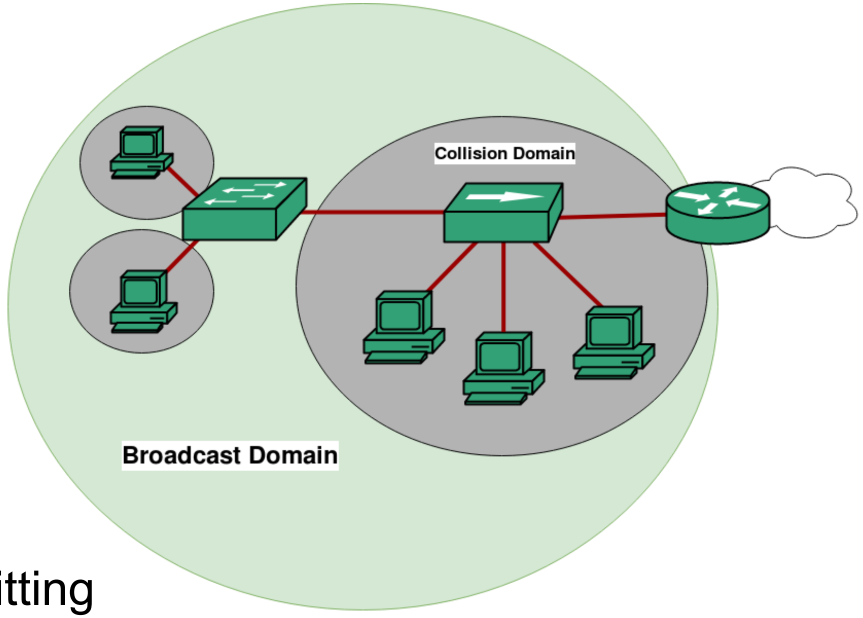
- No mechanism required for tracking whose turn it is
  to access the media
  → Contention-based media access control methods:
  no overhead of controlled access methods

- Possible that two devices or two nodes transmit at the same time
  → = a data **collision**

- Data of both devices → corrupted and need to be resend

Contention-Based Access CSMA/Collision Detection (CSMA/CD)

→ used for Ethernet networks
→ Device looks if the media is free or not.



If a data signal is present
→ media is not free
→ device waits for a random period of time and tries again

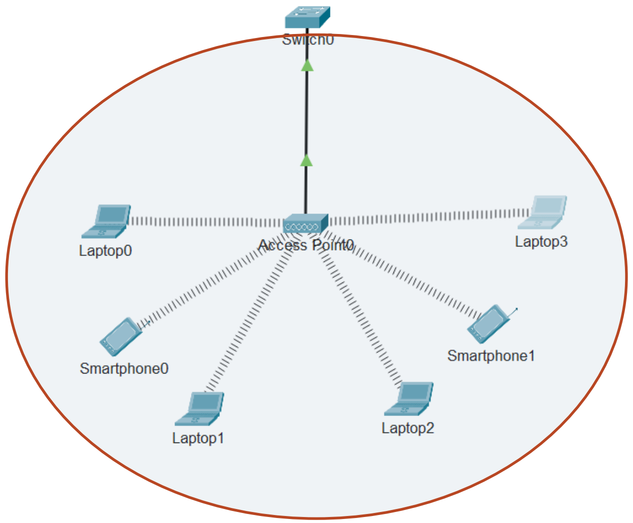
If a data signal is absent
→ media free
→ device transmits the data

→ If a **collision** is **detected** showing that another device was transmitting at the same time
→ all devices stop sending data
→ both devices wait each a random period of time
→ after this period of time (different!) each device looks again if the media is free
→ …

Contention-Based Access
CSMA/Collision Avoidance

→ used for wireless networks

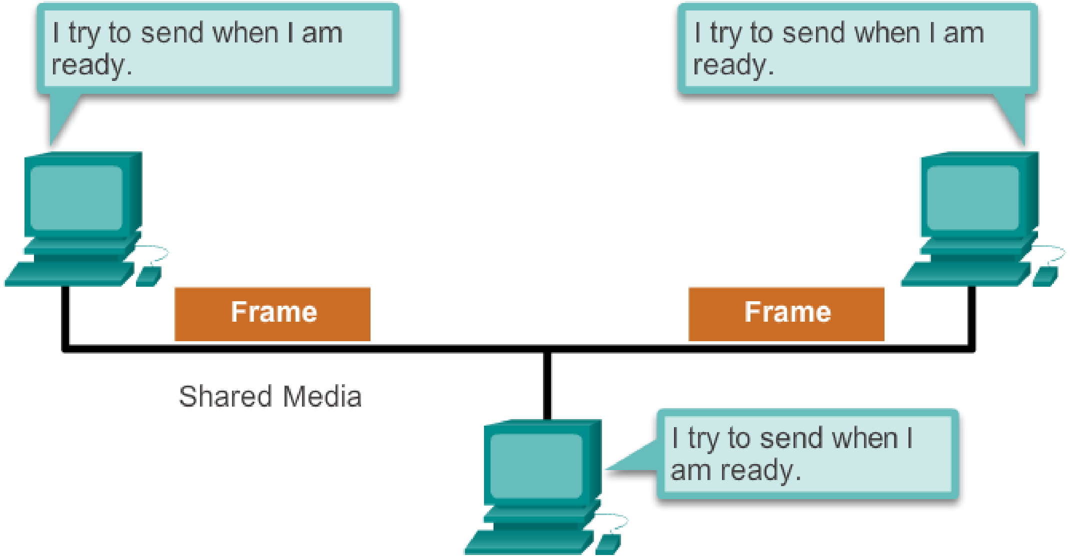
→ Device looks if the media is free or not.



If a data signal is present
→ media is not free
→ device waits for a random period of time and tries again

If a data signal is absent
→ media free
→ device sends a notification across the media to notify everyone that he will send data across the media
→The device then sends the data
→ **Collisions** are **avoided!!**

Contention-Based Access



Method	Characteristics	Example

Contention-Based Access	• Stations can transmit at any time
	• Collisions exist
	• Mechanisms exist to resolve contention problems
	  • CSMA/CD for Ethernet networks
	  • CSMA/CA for 802.11 wireless networks
		• Ethernet
		• Wireless

Power over Ethernet